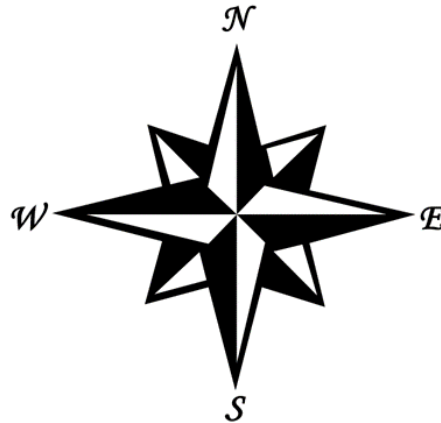


REGION 4 – INTERMOUNTAIN REGION

BOUNDARY MANAGEMENT CONTRACT SPECIFICATIONS



This Region 4 Boundary Management Contract Specifications document contains the specifications and exhibits that will apply to all Forest Service Region 4 Boundary Management and Cadastral Survey Projects.

Please retain this booklet in a permanent file.

All Region 4 Boundary Management, Cadastral and Land Survey solicitations will include the enclosed material, as supplemented for project-specific specifications and exhibits which will be furnished with each particular project.

Contractor:	TBD
Contract Number:	TBD
Project:	Indefinite Delivery Indefinite Quantity A&E Services for Boundary, Cadastral, and Land Surveying Services
Location:	Region 4
Awarding Office:	USDA Forest Service, Business Operations, Procurement and Property Services, Intermountain Zone

Intermountain Region 4
BOUNDARY MANAGEMENT
CONTRACT SPECIFICATIONS

REGION 4 BOUNDARY MANAGEMENT CONTRACT SPECIFICATIONS

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GENERAL SPECIFICATIONS

C-1 SCOPE OF CONTRACT

The intent of this contract is to obtain the land surveying services listed in the schedule of items in accordance with the specifications and exhibits. The Contractor shall furnish all labor, supervision, transportation, equipment, supplies and materials, except the [Government furnished property](#) specified, necessary to complete the work as specified.

All contracted work shall be performed by, or under the direct supervision of, a person licensed to practice surveying in the State(s) where the project is located and be in good standing with the respective State(s) license governing board. The survey shall comply with all applicable State and Local laws and regulations, as well as all applicable portions of the appropriate GLO/BLM Manual of Surveying Instructions and Specifications in this document providing they are not less than these requirements set forth herein.

C-2 DESCRIPTION OF SERVICES

Projects consist of performing all work necessary to properly subdivide and delineate Government Lands as specified in a task order. Such items as corner search and evaluation, control survey and monumentation, property line posting, creation of a Plat, and reports are integral to this project.

The work to be performed under this contract may include, but is not limited to, boundary surveys, retracement and/or dependent resurveys of PLSS rectangular surveys, Homestead Entry surveys, Mineral Surveys, marking and posting of National Forest and Grassland Property Boundaries, Land Claim Surveys, Road/Trail Rights-of-Way surveys, Small Tracts Act and Townsite Act surveys, Special Designated Area surveys, topographic surveys, remote sensing (aerial photogrammetry, bathymetry, etc.), computer aided drafting, miscellaneous services, as well as any other work incidental to the survey profession. Work may also include the maintenance of previously surveyed, marked & posted property lines and corners.

Individual projects will be described in a Scope of Work (or Scope of Contract) for each task order including any supplemental specifications to these specifications and contract Time. Each task order will contain:

Exhibit 1 - Vicinity Map

Exhibit 2 - Corner and Ownership Status

Exhibit 3 - Work to be Accomplished, as well as any other clarifying exhibits as necessary.

A single exhibit may encompass all the information detailed in Exhibits 1-3.

C-3 LOCATION

The contract area includes all portions of National Forest and Grasslands in Region 4: Southern Idaho, Western Wyoming, Utah, Nevada, and a portion of East-Central California. Task orders will be issued when the Government determines its specific surveying needs. Supplemental specifications and exhibits for individual project work sites will be provided with the task orders issued by the Government.

Intermountain Region 4
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DEFINITIONS

Administrative Survey: The line separating areas of differing management objectives, such as Wilderness, Research Areas, Administrative Sites, etc., from areas of general multiple use management.

BLM: Bureau of Land Management.

BLM Manuals: A reference to [BLM publications](#), mainly the appropriate year of the “[2009 Manual of Surveying Instruction](#)”. Also included are BLM reference pamphlets but not limited to, “Restoration of Lost or Obliterated Corners and Subdivision of Sections”, “[Corner Identification and Markings](#)”, “[Durability of Bearing Trees](#)”, etc.

BT: Bearing Tree.

Boundary Line: The line separating lands of the Government from its adjoiners.

CDA: Congressionally Designated Area.

CO: The Contracting Officer has authority to enter into, administer, or terminate contracts and make related determinations and findings.

COR: The Contracting Officer’s Representative is a government official appointed in writing by the contracting officer who provides technical direction, clarification, and guidance with respect to the contract specifications and statement of work.

Contractor: The firm selected by the proponent with the approval of the Forest Service shall hereinafter in these specifications be referred to as the “Contractor.”

Control Survey: A survey which provides the horizontal or vertical position data for the support and control of subordinate boundary surveys.

Coordinate file: A comma-separated-values file consisting of a data table with the minimum of Point Name, Northing, Easting, Elevation, and Description information for each.

DBH: Diameter at breast height, typically measured 4.5 feet above ground on the side of tree facing monument.

Encroachments: Improvements occupied or used on National Forest System land under claim of title or color of title. (36 CFR 254.31)

GLO: General Land Office, precursor to the BLM.

GNSS: Global Navigation Satellite System, including Global Positioning System (GPS - USA) and; GLONASS (Russia), BeiDou (China), Galileo (Europe), QZSS (Japan), IRNSS (India), and any new constellations developed.

Inspector: A person designated by the CO or COR to inspect work performed by the Contractor.

Intervisible Point: A point from which an observer standing on the ground at a post or sign can see the next post or sign on the projected line ahead AND the previous post or sign on the line behind.

PLSS: Public Land Survey System as defined by the BLM.

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Plat: The term “plat” as used in these specifications and exhibits is defined as a Certificate of Survey, Record of Survey, Parcel Map or Subdivision Map in accordance with applicable State law and local ordinances.

ROW: Right of Way.

Record Data: Prior survey data, including but not limited too; deeds, record of surveys, corner cards, parcel maps, lot line adjustments, subdivisions maps, easements, judgements, field verified evidence, unrecorded maps, parol evidence, exhibits, BLM/GLO furnished data, FS furnished data, etc.

Tie Line: A survey connection from a point or corner on a boundary line to a point whose position is required, or a connection to close a survey on a previously determined point.

Trespass: To go on another's land or property unlawfully; an illegal act causing injury to the person or to the relative rights of another's property; also, legal action for damages resulting from trespass (FSH 5509.11, Zero Code). Trespass includes constructing, placing, or maintaining any kind of road, trail, structure, fence, enclosure, communication equipment, significant surface disturbance, or other improvement on National Forest System lands or facilities without a special-use authorization, contract, or approved operating plan when such authorization is required (36 CFR 261.10).

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C-4 MATERIALS

A. Government Furnished Property

All Government-furnished property will be provided as needed at a location specified by the CO or applicable COR.

The Government shall furnish the following listed items:

1. All signs required by the Specifications (see [Exhibit .424](#)).
 - a. 54-2 Property Boundary Signs or Decal Stickers
 - b. 54-3 Bearing Tree Signs
 - c. 54-6 Location Tags
 - d. 54-9 Land Survey Monument Signs or Decal Stickers
2. Monuments.
 - a. Iron, Steel, or aluminum pipes with brass, steel, or aluminum caps.
 - b. Brass or aluminum tablets suitable for placement in concrete or stone.
 - c. Aluminum caps for use with 5/8" 'rebar' or other drive type monuments.
3. Metal delineator posts or red steel angle iron posts, or fiberglass marker posts, as provided.
4. Underground magnetic markers.
5. Tree marking paint.
6. Forest Service supplied data including but not limited to corner cards, and field notes and plats from prior surveys. This data may not be accurate or complete and shall be considered only as informational. The Contractor shall be responsible for its acceptability.

All Government-furnished property will be provided as needed at a location specified by the CO or the COR. The Contractor shall give a minimum of 48 hours prior notice to the materials delivery. Deliveries will be made only during the Government's normal business hours.

B. Contractor Furnished Materials

The Contractor shall furnish all materials required to accomplish the work, except for the Government-furnished materials (Refer to Section [C-4 A.](#)).

At the termination of the contract, all items branded Forest Service (signs, monuments, etc.) and posts shall be returned to the COR.

C. Materials Specifications

1. 16D, 3 inches or longer aluminum nails shall be used for fastening signs to trees and shall be of such length/size as necessary to penetrate firmly into the tree's sapwood/heartwood layer leaving a 1/2" of nail length exposed to allow for tree growth ([Exhibit .423](#)). Aluminum nails must be used as a safety factor in case the tree is ever taken to a sawmill or cut with a chainsaw.
2. Control traverse points – 6 inch 60D nail or larger steel spike.
3. Guard stake lath – 4 feet x 3/8 inch x 1-1/2 inches or larger if needed.
4. Flagging – Pink or as specified.

C-5 TECHNICAL SPECIFICATIONS – CADASTRAL SURVEYS

All surveys necessary to define National Forest boundaries shall comply with applicable State laws and regulations. Where State laws and regulations are less restrictive than the specifications contained herein, these specifications shall apply. In Public Land Survey System (PLSS) States, the principles and practices as set forth in the BLM Manuals, as well as applicable State laws, shall be followed. Where these specifications differ from the Manuals, these specifications shall apply.

.100 RESEARCH

.110 Record Research

It is the Contractor's responsibility to research the real property records of public agencies to obtain the title and survey history of the subject property and relevant adjoining properties. The Forest Service shall provide the Contractor access to National Forest System landownership status, corner search and survey files at the time of award.

Errors & significant inaccuracies pertaining to title and boundary found in Government-supplied records shall promptly be reported to the COR. However, any information given to the Contractor by the Government shall be construed as being for informational purpose only, and the sole responsibility for its accuracy, completeness, and/or verification rests with the Contractor.

.120 Adjoiner Notification

Prior to the start of field work, the Contractor shall contact affected adjoiners of Government land by mail to secure permission to access/enter their lands to survey, mark accessories, and mark and post boundary lines. Some State Laws may require certain statements to be included in these contact letters. The Contractor shall furnish the Government documentation of such contacts listed in the following categories: Adjoiners who have granted permission; those who have granted conditional permission, specifying conditions or restrictions; those who have denied permission; and those who are non-responsive within 14 calendar days of attempted contact. The format of a written request and reply card shall be similar to that shown in [Exhibit .120](#).

Should permission for access be denied, the Contractor shall notify the COR immediately and relay such information giving names, dates, and text of conversations which have taken place between the landowner(s) and the Contractor. For those areas of the boundary line where permission has been denied, the Government will then notify the Contractor as to what portions of the work will not be performed along those lines.

.130 Conflicts, Boundary and Title Disputes

The Government will inform the Contractor of any known or suspected conflicts of survey, possession, occupation, or title along the boundary lines to be surveyed. Any conflicts discovered by the Contractor during the course of the project shall be reported to the COR as soon as practical, followed within five (5) days after verification with written documentation in a format like that in [Exhibit .130](#), see [.920 Encroachment Diagram](#).

The Contractor is not required to resolve title or possession conflicts but is required to report facts and any professional opinions that may be relative to the conflict. For all surveys, the Contractor shall be prepared to testify in court, if called upon by the U.S. Attorney. In consideration of the performance of such services, payment will be made by the Department of Justice in accordance with its prescribed regulations and procedures.

An equitable adjustment of contract performance time or contract price may be made for any addition or deletion of survey work authorized by the Contracting Officer resulting from a conflict. If the conflict is resolved within the contract time limits, the Contractor shall mark the disputed boundary line and corners as specified.

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.140 Technical Advice

All requests submitted to the Bureau of Land Management (BLM) for technical advice on survey matters shall be in written form. The request shall be sent directly to the BLM with copies sent to the COR. The letter shall include the following:

1. Contractor name.
2. Name of COR.
3. cc's indicating copies have been sent to the CO and COR.

The request shall be routed through the COR if the request involves a dispute or conflict between private surveyors. A request for technical advice shall be sent by the COR in cases of un-resolvable differences of opinion between the COR and Contractor regarding survey matters incident to the project. The BLM's formal response shall be adhered to by the Contractor.

.200 CORNER SEARCH, EVALUATION AND DOCUMENTATION

The Contractor shall make an investigation of all corners as indicated on Exhibit 3. Every corner used as a part of the survey must be independently verified by or under the direct supervision of the licensed land surveyor in responsible charge as to authenticity of corner evidence and integrity of the monumented position.

.210 Field Search

The Contractor shall investigate and search for monuments and other physical evidence as indicated on Exhibit 3 and included in Section B – SERVICES AND PRICES. An extensive on-the-ground search of the project area shall be conducted to obtain evidence left by previous surveys, including original monuments and accessories, such as bearing trees, line trees, and reference objects. When searching and evaluating the physical evidence of the original corner monument and accessories, the utmost professional care shall be taken to protect and preserve the evidence.

One of three difficulty levels is identified for each corner, or the whole project:

Level 1 - Difficult Applies to recovery of corners originally set with stone or wood post monuments, or later surveys with extensive obliteration as determined by the COR.

Level 2 - Normal Applies to brass-cap era surveys.

Level 3 - Easy As determined by the COR, corner search may not be applicable if the survey is a recent (10 years or less) aluminum/brass cap survey *or* if corners have been recovered by the USDA Forest Service. Corner documentation is still required.

.220 Corner Documentation

When a corner has been found, monumented, or determined to be lost or obliterated, the Contractor shall state this in the field book, the prescribed State/county corner record form (see [Exhibit .430](#)), and/or a Record of Survey, as directed by the COR. The State/county corner record form shall be used following applicable State rules and regulations, or form FS-7100-52 ([Exhibit .220](#)) for Utah or other States without a prescribed form. In the recorded document the Contractor shall address each corner monument or accessory of prior record and explain the rationale used to harmonize conflicting evidence, if any, in sufficient detail to allow another competent surveyor to examine and accept the solution without consulting additional records.

All PLSS corners noted in the field notes and documents shall be referred to and numbered or named by BLM's Geographic Coordinate Data Base (GCDB) System numbering scheme.

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.230 Corner Reporting

All corner records shall be furnished to the COR and be submitted in the manner as required by the State in which the survey is performed. In addition, a CSV file containing corner State Plane coordinates shall be provided to the COR.

.300 CONTROL SURVEY, FIELD DATA COLLECTION AND COMPUTATIONS

.310 Survey Methods and Procedures

The Contractor shall select the methods and instruments to be used for field data collection and boundary location. Methods such as photogrammetry, traverse, triangulation, GPS/GNSS systems may be used. The system selected shall meet the positional tolerance requirements described in the specifications.

.311 The Position of Each Corner

The position of each corner established or tied shall be determined by a closed traverse loop, redundant radial survey ties and/or point positioning procedure designed to meet the standards in [.320 Survey Standard](#).

.312 Determination of Bearing

All bearings shall be referenced to a published coordinate system and datum, with the method of determination noted in the field notes and plat. A Record Basis of Bearing is acceptable when referenced to an official Government or Private survey of Public Record. A GPS Basis of Bearing shall be established as described in [.321 Global Navigation Satellite System](#). If compass bearings are used for corner accessories, the Contractor shall include the declination used and method of determination. Use of assumed bearings is not acceptable.

.313 Determination of Distance

All distance measuring equipment used in conjunction with this contract shall be calibrated annually or per manufacturers recommendation (whichever is more frequent) and results given to the COR upon request.

.314 Marking the Survey Point

Survey control points shall be marked with the minimum of a 60D nail or larger firmly anchored and identified with appropriate point number written on attached flagging, as a minimum. When conditions warrant, a guard stake or lath shall be placed alongside the point with identification.

.315 Clearing of Survey Lines

No trees larger than 6 inches DBH shall be cleared along survey control ties (not along boundary) without the permission from the CO/COR. Clearing shall not be performed on private property without prior permission of the landowner. Under no circumstance shall fruit, shade or ornamental trees, shrubs, or hedges in residential or developed areas be blazed, marked, or severed. No cut trees shall be left leaning against or hanging in standing trees.

.316 Boundary Line Survey Ties

The Contractor shall locate and tie the following items to the control survey system:

1. All corner monuments including accessories, cutting boundary monuments and other evidence indicating a previous survey, whether recorded or not, along the boundary.
2. Visible signs of possession or encroachment such as structural improvements, limits of timber cutting, mining operations, fences, plowed fields, etc.
3. Major topographic or cultural features such as highways, roads, railroads, trails, and rivers crossing the boundary line.

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All items located will be shown on either; the survey map exhibit, plat, or trespass/encroachment diagram. The document showing these items will be approved by the COR and may be in any combination of the deliverable documents.

.320 Survey Standard

The positional accuracy as published by the National Geodetic Society (NGS) is the radius of a circle or an ellipse of uncertainty, such that the true or theoretical location of the point falls within that circle or ellipse 95% of the time (1.96σ).

- Project control and/or control for new monumentation shall be no less than Third-Order, Class I (1:10,000) with a linear error of closure not exceeding 2.0 feet before angular adjustment.
- Surveys for marking and posting shall be no less than Third-Order, Class II (1:5,000).

A closure or processing report may be requested by the COR at any time and shall be supplied by the Contractor.

.321 Global Navigation Satellite System (GNSS)

Geodetic/survey grade GNSS equipment and methods may be utilized for measurements when it saves cost and/or time compared to conventional methods and the forest tree canopy is such to allow its use. Using multiple satellite systems (GPS, GLONASS, Galileo, or BeiDou) capable equipment may significantly increase satellite density, accuracy, and production in heavy vegetation. At least one control corner for projects less than 1.5 miles in width and two minimum for larger projects shall be directly geo-referenced for entry into the corporate GIS database.

If GNSS is used, the surveyor must follow the “[Standards for Positional Accuracy for Cadastral Surveys Conducted Using GNSS](#)” AND “[Guidelines for the Use of Global Navigation Satellite Systems \(GNSS\) in Cadastral Surveys](#)” prepared by the United States Department of Agriculture – Forest Service and the United States Department of the Interior – Bureau of Land Management.

Local Accuracy Standards per Guidelines shall be used:

<u>Application</u>	<u>Semi-major axis 95 percent error ellipse</u>
Cadastral Project Control	Less than 0.025 (m)
Cadastral Corner Measurements	Less than 0.050 (m)

Caution: The use of GPS does not relieve the Contractor from the responsibility to search and evaluate all evidence of original corners and/or subsequent surveys along a surveyed line; to tie evidence of encroachment and occupation onto Government or adjacent land; and to set sign posts as outlined in [Exhibit .521](#).

.330 Field Notes

Field notes shall be recorded in writing, in a standard waterproof field book, and/or by electronic data collectors. The method used shall provide a complete and accurate record of all field work.

.331 Field Books

Errors shall be lined out and not erased. Legible photocopies (may be in digital format) of the original field books shall be furnished to the Government upon completion of the project.

Field books shall contain, at a minimum, the following data:

1. The contract and work order number and name shall be printed in ink on the cover of each book and each book shall be sequentially numbered.
2. Each book shall contain a small sketch and index page in the front of the book which will include a brief description of work and the corresponding page number.

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3. Names and duties of each party member, weather conditions, instruments used, PPM or other correction for EDM, the location of the work performed, and the date shall be recorded each day work is performed.
4. Field data collection and boundary layout notes shall be recorded in such a fashion as to allow another technically qualified person to retrace the survey without consultation with the Contractor.
5. Basis of bearing data shall be recorded in sufficient detail to permit an independent check.
6. Statements by adjoining or interested parties relative to boundary line and/or corner locations.
7. Field record of all corners found or monumented with a description of the monument and all accessories and remains of accessories. Descriptions shall include bearing, distance, and DBH measurements.
8. Surveyor's Certificate: The following certification is to be made immediately following the last page used in the notebook:

I hereby certify that the foregoing notes represent work done under my direct supervision.

Date

Signature

State Registration No.
and Seal

Title

.332 Electronic Field Book

An electronic copy of the field notes will be provided in an agreed upon format (PDF, DOC, TXT, etc.) at the end of the task order. The COR may request the document in its current form at any time during the course of the survey.

Field Notes shall contain, at a minimum, the following data:

1. The contract and work order number and name shall be listed in the title and header, sheet number with document overall number of pages to be added to the footer.
2. Names and duties of each party member, weather conditions, instruments used, PPM or other correction for EDM, the location of the work performed, and the date shall be recorded each day work is performed.
3. Field data collection and boundary layout notes shall be recorded in such a fashion as to allow another technically qualified person to retrace the survey without consultation with the Contractor.
4. Basis of bearing data shall be recorded in sufficient detail to permit an independent check.
5. Statements by adjoining or interested parties relative to boundary line and/or corner locations.
6. Field record of all corners found or monumented with a description of the monument and all accessories and remains of accessories. Descriptions shall include bearing, distance, and DBH measurements.
7. Surveyor's Certificate: The certification is to be made immediately following the last page used in the notebook and the affixed seal and signature can be digital in nature.

I hereby certify that the foregoing notes represent work done under my direct supervision.

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.333 Electronic Data Collectors

If the Contractor uses an electronic data collector, the following shall be furnished to the Government:

1. An electronic copy containing the original field data prior to any manipulation or adjustments.
2. A separate written report showing:
 - a) The data format used, describing the data on the hard copy by line and column.
 - b) All descriptive codes and their meaning.
 - c) A small sketch of the project with an index showing the data contained on each type of media.

.334 Legible Copies

Legible copies of all field books used in conjunction with the collector which describe work done, including but not limited to a complete description of all monumentation found, used, or set during the survey shall be provided by the Contractor.

.340 Survey Control Diagram and Computations

A survey control diagram and a copy of all computations shall be delivered as requested by the COR.

.341 Survey Control Diagram

The Contractor shall provide a survey map exhibit to the COR upon request as soon as practical. The Contractor shall furnish the Government with a digital copy (or paper upon request) of the survey in its most current form. The size and file format of each exhibit shall be agreed upon with the CO/COR and should contain the following information.

1. Project name, contract number and date of exhibit.
2. Contractor's name or business name.
3. Project location including Township, Range and Base Meridian.
4. Dates of each revision.
5. Linear scale (showing sufficient detail).
6. Basis of Bearing, coordinate system, and datum.
7. Survey control stations and monuments tied or set, accurately plotted with names or numbers.
8. Raw and final coordinate values and all boundary/control lines measurements between all corner monuments used, established, or re-established.
9. BLM Geographic Coordinate Data Base (GCDB) identifiers and reliability codes for all corner monuments.
10. References to survey field notes and all compiled record data ([Definitions](#)).
11. Tie and boundary lines shall be shown on the control diagram.

.342 Computations

A legible copy of all computations shall be provided if requested by the Government in AutoCAD .DWG format by method agreed upon by CO/COR. Computations shall list survey control point names or numbers, survey field data and an applicable adjustment report. The adjustment report shall state the type of adjustment used (compass rule, least squares, etc.) including any applicable data and both raw and adjusted coordinate values for survey points and corner ties.

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.400 CORNER MONUMENTATION, PERPETUATION, AND ACCESSORIES

.410 Monumentation

Monumentation shall comply with the BLM Manuals and specific State laws, and be in accordance with [Exhibit .411](#), unless otherwise directed by the COR. The COR shall provide direction to the Contractor on a case-by-case basis.

The Government will supply monuments for all Government boundary and Public Land Survey corners to be monumented or re-monumented by the Contractor as part of the survey (See [C-4 A. Government Furnished Property](#)). Magnetic memorials or suitable ferrous material such as a rebar iron spike shall be placed at the base of new pipe monuments, unless otherwise directed by the COR. Any corner position which cannot be monumented in its true location shall be referenced by witness corners or reference monuments only after a thorough review of [Exhibit .412](#).

1. ***Difficult*** - Dig in, Using a standard USDA Forest Service 2 ½" x 30" Steel or aluminum pipes with brass or aluminum caps., or rock tablet.
2. ***Easy*** - Drive in, Using a drive rod or rebar and cap monument.

.411 New Monumentation Placement

Monuments shall be placed as illustrated in [Exhibit .411](#). Physical evidence of earlier surveys should be preserved when possible. Two clear color photographs (preferably digital) shall be taken of each original monument in place before disturbance. The Contractor shall bury original stones upside down in the hole alongside the new monument and place original wood posts near the new monument for inspection by COR. Brush and logging debris within a radius of 6 feet of the corner shall be cleared away when the monument is set. Line of sight to each bearing object existing or established shall also be cleared.

Monuments set by this survey shall be located to the tolerance requirements specified in Section [.310 Survey Methods and Procedures](#). Placement shall be in accordance with the BLM Manuals.

New monuments shall be solidly placed in earth ([Exhibit .411](#)) when possible. A magnetic memorial shall be placed at the base of new pipe monuments, unless otherwise directed by the COR. If applicable, the Contractor should bury original monuments alongside the new monument. This information shall be recorded in the field book and documented in the pertinent Corner Record or Plat.

.412 Reference and Witness Monuments

Where it is impossible or impractical to place a monument at the true corner position, reference monuments and witness corner monuments shall be established as prescribed by the current version of the BLM Manuals. See [Exhibit .412](#) for examples.

.413 Monument Cap and Tablet Markings

Monument caps and tablets shall be stamped as prescribed in the current version of the BLM Manuals. The caps or tablets shall also be stamped to include the year of survey and the surveyor's license number. If applicable, the stamping shall include any unique Forest Service tract or corner identification. If under subsequent review by the COR the marking is determined to be incorrect, the monument shall be re-stamped at the Contractor's expense.

.414 Corner Trees

When the corner position is coincident with a live, healthy tree, the tree shall be marked as prescribed in the current version of the BLM Manuals, or as directed by the COR.

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Corner trees shall be perpetuated with a reference monument which shall count as one of the required accessories. The monument shall be set a minimum of 15 feet from the base of the corner tree.

.415 Original GLO Line Trees

Positively identified original Government Public Land Survey line trees typically set by the GLO are monuments of the true property line (not to be confused with modern line trees set per other surveys such as those outlined in these specifications). When recovered in good condition the marks shall be painted and sign "Land Survey Monument", or equivalent Government sign substituted by the COR, shall be attached in accordance with [Exhibit .424](#). When recovered in down and/or decayed condition the true position shall be re-monumented.

See BLM manual 6-28, perpetuating line tree 4-113 (4); When an accessory is necessary to evidence the position of a line tree, no requirements are set up other than to mark a bearing tree or a bearing object, if available, or to establish a suitable bearing object. A bearing tree is marked in the same manner as a bearing tree associated with an angle point.

.420 Bearing Trees and Other Accessories

Refer to the BLM Manuals for selection, measurement and marking of bearing trees and other objects. Limbs on bearing trees shall be removed to a height of 7 feet.

.421 Selection

All PLSS corners shall be selected according to the BLM Manuals and referenced by at least two accessories, or more as specified by applicable State laws.

The Contractor shall use "Durability of Bearing Trees" as a guide when selecting bearing trees and shall take extra care to select trees that are healthy, free of defective tops, lack missing bark, and middle-age or younger whenever possible.

.422 Measurements

All bearing trees shall be described by bearing and distance according to the BLM Manuals.

.423 Blazing, Scribing and Painting

Blazing, scribing, and painting shall be in accordance with the BLM Manuals and [Exhibit .423](#).

.424 Signing

Available signs with identifying numbers are illustrated by [Exhibit .424](#).

1. Bearing Tree signing shall be in accordance with [Exhibit .423](#). Sign 54-3, "Bearing Tree," shall be placed on all newly created bearing trees and on all existing bearing tree remnants.
2. Sign 54-9 "Land Survey Monument" shall be placed on the back of all standing bearing trees - dead or alive.
3. A guard post with sign 54-9 or P71-10 decal attached shall be placed at all corner monuments in accordance with [Exhibit .521](#).
4. If bearing trees not available and monument falls within close proximity of a fence, the appropriate signs shall be placed on fence posts. (Sign 54-5 for rectangular PLSS corners, Sign 54-6 for section subdivisions and 54-8 signs for Homestead Entry and Mineral Surveys).

.430 Corner Recordation

If required by State statute, the Contractor shall complete, in printed and/or electronic form, as specified by the Forest Service in accordance with State and local statutes, a Corner Recordation Form (or local equivalent) for

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each corner required by Exhibit 3 or by State and local statutes. After review by the COR, the Contractor shall file the original in the appropriate jurisdiction. A copy of the recorded form with completed filing information shall be furnished to the Forest Service. Refer to [Exhibit .430](#) for completion guidelines and sample of a completed form.

An original completed form FS-7100-52 may be required for the task order as requested by Exhibit 3. All information on the card shall be completed in accordance with [Exhibit .220](#).

.500 BOUNDARY LINE LOCATION, MARKING AND POSTING

The boundary lines to be marked, shown on Exhibit 3, shall be located by survey, unless work is to be performed as “Boundary Line Maintenance” in accordance with [.700 CORNER MAINTENANCE](#). The methods used shall ensure that posts, blazes on trees, and/or line monuments defining the boundary line have a positional tolerance meeting the standards of [.310 Survey Methods and Procedures](#).

The COR may direct the Contractor not to perform certain work required under this section when marking lines in developed areas and areas near homes. This may include not painting blazes, increasing the blazing spacing, installing different kinds of signs or posts, increasing the post/sign spacing, or other measures.

.510 Locating Boundary Lines

Boundary lines shall be marked by installing steel, aluminum, or fiberglass posts within ± 1 foot of the true line. This specification shall not be interpreted as allowing posts or line monuments to be placed off the true line unless otherwise directed by the COR.

.520 Marking the Property Line

Property lines are marked by placing line posts, line monuments, painted blazes on trees, and clearing. The Task order will specify one of the following levels of intensity per mile or per the whole task order. Markings on original Government Survey trees (corner, witness, line, etc.) shall not be destroyed.

1. **Intensity Level 1:** 100 percent of the boundary will be marked per this section. Any deviations from this requirement must be approved by the COR or shall be addressed in Task Order Scope of Work if changes are requested by the Government regarding blazing and painting of trees.
2. **Intensity Level 2:** Only the property line within 50 feet either side of a boundary sign and on Forest lands within 3 feet of the property line shall be marked with painted blazes on trees and within 2 feet of the property line be cleared per this section. Any deviations from this requirement must be approved by the COR or shall be addressed in Task Order Scope of Work if changes are requested by the Government regarding blazing and painting of trees.
3. **Intensity Level 3:** Open areas and Grasslands, posting and/or line monuments only, no blazing or clearing required.

.521 Placing objects to mark the property line

Posts, line monuments and line blazes shall be placed only on the true property line located to the standard specified above. Posts shall be set as per [Exhibit .521](#), without deforming or degrading the usefulness of the post.

.522 Blazes

Blaze shall be made in accordance with [Exhibit .522](#). All trees 6 inches DBH or larger and within 3 feet of boundary lines shall be marked with the appropriate blaze. Existing blazes found in the correct location and healed over shall not be reblazed, but shall be painted. Blazes shall be cut through the cambium layer. Blazing

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on the adjoiner's side of the line shall not be performed if the adjoiner denies permission to mark their side of the line.

.523 Clearing

Within 2 feet of each side of the boundary line, the Contractor shall clear line sufficient to maintain intervisibility of posts and remove trees less than 6 inches DBH as allowed by permission from the adjoining landowners. The resulting debris from the clearing shall be lopped to settle to within 1 foot of the ground. All limbs shall be cut off the stumps of felled trees to prevent regrowth. Stump height shall not exceed 6 inches. Lopping stems in a criss-cross pattern, resulting in hazardous pointed spikes shall be avoided. Cut trees shall not be left leaning against or hanging in standing trees unless otherwise directed by the COR. Vegetative debris shall be scattered to reduce concentrations, deposited offline, and left on the Government side whenever possible. Debris shall not be deposited within 5 feet of roads, trails, stream courses or fire breaks.

Exceptions to this clearing standard in certain areas of heavy chaparral or close proximity to occupied dwellings may be approved at the discretion of the CO/COR. Any equitable adjustments in unit price for reduced clearing will be negotiated with the Contracting Officer.

.524 Painting

All blazes applied under .522 Blazes shall be painted red in accordance with [Exhibit .522](#). Paint shall be supplied by the Government. Paint placed by the Contractor in an improper location shall be corrected, at the Contractor's expense, by painting over with black paint supplied by the Government.

.530 Posting the Property Line

Posting is the act of placing signs and posts along the property line, so it is recognizable. Placement of signs along the boundary line shall be in accordance with [Exhibit .521](#). Posts will be placed at intervals not to exceed 200 feet, or shorter to remain inter-visible if brush and trees were removed, (signs on line trees do not satisfy the interval spacing, only sign posts). Some task orders may adjust this distance depending on vegetation and the priority of the project. Some of the signs that are available are shown in [Exhibit .424](#). Other signs may be specified for special projects.

.531 Post and Sign Placement

Placement of posts along the boundary line shall be in accordance with [Exhibit .521](#) and [Exhibit .522](#).

When choosing post locations, consideration shall be given to the visibility of the post and sign from off line, e.g., a post shall not be set immediately next to and facing a tree that will block its visibility and shall be moved along the true line until it is in a position from which it can be seen when standing off line. Posts shall also be placed at ridge, stream, road, and trail crossings. Pink flagging shall be attached to the top of all sign posts. "Property Boundary" signs or stickers shall face out/away from Government lands, and shall be aligned parallel with the boundary line.

The first sign post along the property boundary (referred to as a guard post) shall be explicitly on the boundary line and placed ± 5 feet from the corner and have a "Land Survey Monument" sign or sticker, or equivalent Government sign substituted by the COR, attached. The second post on line out of the corner shall be placed 50 to 75 feet from and be visible from the corner, then normal sign pacing should continue per [.530 Posting the Property Line](#) and [Exhibit .521](#).

Posts shall be driven in the ground a minimum of 18" and be securely planted (with no side-to-side play and difficult to remove). If this cannot be achieved a rock mound shall be built around the post to 24", or higher, until secure. Rock (and not wood or other organic material) is the only material that may be used to provide

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supplementary support to posts that are not securely planted after being driven to refusal. The width of rock mounds shall be twice their height.

All signs and posts placed in improper location shall be removed or relocated at the Contractor's expense.

.532 Property Boundary Sign, 54-2 or 54-7; Decal P54-1 or P54-2

Property Boundary Decals or signs shall be placed on line posts or trees on the boundary line. Posts shall be placed at a maximum of 200 ft. from each other. Posts or signs on trees shall be placed at ridge, stream, road, and trail crossings intervisible from each other. Where trails and roads are intersected, posts shall be placed on both sides of the road/trail, no less than 10 feet from each side of the traveled way and no more than 25 feet on one side and 50 feet on the opposite side.

See Special Project Specification for spacing and visibility requirements between signs. These signs shall not be placed on private trees.

.533 Location Poster Sign 54-5, 54-6, or 54-8

*Location posters are not required per this contract, but it may be requested for specific task orders. If so, the following shall apply:

The location posters shall be placed where all marked boundary lines cross roads and trails. Sign 54-5 is for section and township lines. Sign 54-6 is for section subdivisional lines. Sign 54-8 is for metes and bounds survey lines. These signs may be placed on either posts or trees. The Contractor shall stamp or etch the required information on the sign. Some additional special posts at road crossings may be required. When required, they shall be shown on Exhibit 3 and included in the Special Project Specifications.

.534 Sign Attachment

The Contractor shall firmly attach signs and/or stickers to posts and appropriate line trees in accordance with [Exhibit .521](#), [Exhibit .522](#), and Special Project Specifications. If a fence consisting of wire and steel fence posts is within 1 ft. of line - with the landowner's permission - signs shall be attached to the top wires with hog rings at the four corners of the sign and placed next to the steel fence post. If a fence consists of wood fence posts and wire, the sign shall be wrapped around the fence post and nailed to the post at all four corners of the sign with galvanized nails. See Special Project Specification for spacing and visibility requirements between signs. The Contractor shall firmly attach signs to sign posts with corrosion resistant bolts and nuts, aluminum rivets, or corrosion resistant wire as directed by the COR.

.540 Obliteration of Previously Posted Lines

After some land acquisitions or disposals, the Government may order the obliteration of previously posted lines. Government signs and posts previously placed which are no longer within one foot of a boundary line shall be removed from the site by the Contractor and shall be delivered to the Government. Old paint on the Government side of previous boundary lines shall be painted over with black paint provided by the Government.

The Contractor shall notify the CO/COR upon the discovery of Government signs and posts previously placed which are not on the boundary line. Separate task orders will be issued for the removal of posts and signs as required. When deemed by the CO/COR, the Contractor shall remove these signs and/or sign posts. They shall remain the property of the Forest Service.

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.600 PLATS

The Contractor shall prepare a plat(s) in accordance with Section B - SERVICES AND PRICES - of this contract which meets the State law and county requirements similar to that illustrated by [Exhibit .600](#). Prior to filing of the plat, the computations may be requested for review.

The Government's acceptance of the plat does not release the Contractor of any liabilities through errors or omissions.

.610 Content

All appropriate information contained in the Cadastral Plat Check List (see [Exhibit .600](#)) shall be included on the plat.

.620 Review

Prior to filing, all items on the plat checklist contained in [Exhibit .620](#) shall be reviewed and initialed by the Contractor and submitted with one paper copy of the plat to the COR for review.

.630 Geodetic Position

Plats for cadastral projects shall show a geodetic position on a minimum of two corners. A statement shall be placed on the plat as to the method used to determine the position, the datum and the date established. See [.321 Global Navigation Satellite System \(GNSS\)](#) for accuracy standards. Geodetic positions shall be provided on the Control Diagram if a plat is not prepared.

.640 Filing or Recording of Plats

The final plat(s) shall be submitted to the COR for review prior to filing or submitting to the local review authorities. After the required reviews, the Contractor shall be instructed to file the plat(s) and furnish the required copies to the Forest Service as follows:

1. Recorded copies: The Contractor shall file or record the survey plat(s) in the appropriate jurisdiction in accordance with the State and Local Statutes. Filing fees and any other costs involved shall be borne by the Contractor. After acceptance and recordation, a digital copy of the recorded document shall be furnished by the Contractor to the Government in the format provided by the county.
2. Forest Service copies: If requested in the task order, a reproducible copy of the final plat, consisting of a permanent image on a stable base polyester (mylar), with proof of filing, shall be furnished to the Forest Service.

.650 Digital Copy of Plat

A digital copy of the prepared plat shall be given to the Forest Service in PDF or TIFF format. In addition, a CAD file (e.g. AutoCAD .DWG format) shall be submitted in a media as requested by the COR.

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TECHNICAL SPECIFICATIONS – MAINTENANCE

Maintenance is the act of bringing previously surveyed, marked & posted boundary lines and corner monumentation of various conditions back to current standards. This infrastructure can degrade over time from environmental and human factors, among others. While it is not required, use of full GNSS survey systems can expedite the maintenance process. The following [.700 CORNER MAINTENANCE](#) and/or [.800 BOUNDARY LINE MAINTENANCE](#) operations shall be performed at each corner or line indicated.

For projects of maintenance only, adjoining landowners shall be notified, see [.120 Adjoiner Notification](#). Refer to [Exhibit .120](#) for example notification letter.

.700 CORNER MAINTENANCE

The Contractor shall perform the following work at the previously monumented corners identified on Exhibit 3 of the task order. Original markings on trees (corner, bearing, witness, line, etc.) shall not be destroyed. Refer to [.400 Corner Monumentation, Perpetuation, and Accessories](#).

.710 Monument Maintenance

.711 Brushing

Brush and debris within a 6-foot radius of the corner shall be cleared away. Care shall be exercised to avoid destroying the original monument or accessory evidence during this operation.

.712 Existing monuments

The Contractor shall ensure that the corner monument is firmly and securely set in place in the ground. Loosely set or badly tilted monuments shall be reset. Scattered mounds shall be rebuilt and corners without mounds and with rock available within a radius of 100 feet shall require that a mound be constructed similar to that illustrated in [Exhibit .411](#).

.713 Resetting monuments

The Contractor shall notify the COR of any damaged, missing, or out of place monuments. After notification the COR may, at their discretion and with the approval of the CO, direct the Contractor to re-set the monument either by Control Survey or from BTs. In such cases an equitable adjustment of contract time and/or price will be authorized by the CO for the additional work needed to re-set the monument.

.720 Bearing Trees and Accessories Maintenance

A search for each BT and/or accessory of record shall be performed by the Contractor and shall be maintained as follows:

.721 Visibility, Painting, and Signing

A search for each BT and/or accessory of record shall be performed by the Contractor. All evidence found shall be made visible from the corner, painted, and signed in accordance with [.423 Blazing, Scribing and Painting](#) and [.424 Signing](#).

.722 Establishment of New Accessories

The Contractor shall establish new BTs or objects in accordance with [.420 Bearing Trees and Other Accessories](#) to replace dead, damaged, dying, or missing BTs or bearing objects and for the number and location (quadrant placement) of healthy BTs required at a corner. All new BTs shall be visible from the corner. A State or county corner record shall be filed if required by state statute, and a copy provided to the COR.

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.730 Selection, Measurements, Blazing, Painting, Scribing and Signing

Refer to [.420 Bearing Trees and Other Accessories](#).

.740 Corner Recordation

Refer to [.430 Corner Recordation](#).

If a corner record or FS corner card is not ordered for maintained monuments, then field notes for each corner maintained shall be recorded in the format specified in [.330 Field Notes](#) and be submitted to the COR when requested. Field notes shall contain, at a minimum:

1. Description of the monument found and any other evidence of the corner point.
2. Sketch or picture clearly showing the markings on the monument cap.
3. Description of all BTs, BT remains, and other accessories found, including DBH
4. Description of all new BTs and/or accessories established, including DBH
5. Bearings and distances measured from the corner to all existing and new BTs, BT remains, and other accessories. The basis of bearing and declination used for such measurements shall be noted.
6. Date work was performed, and equipment used to perform measurements, such as hand compass, EDM, steel tape, total station, staff compass, GNSS offset, etc.
7. Description of any other work done to maintain and preserve the corner monument and/or point.

.750 Corner Coordinates

The Contractor shall provide resource grade coordinates, at a minimum, for all corners maintained.

.800 BOUNDARY LINE MAINTENANCE

.810 Scope of Boundary Maintenance

The Contractor shall perform all work necessary to bring the property lines identified for maintenance on Exhibit 3 into compliance with the specifications in this section. The boundary line should be obvious from previous marking and posting. If the line is not obvious, the Contractor should immediately notify the COR. Original markings on line trees (corner, witness, line, etc.) shall not be destroyed. Work on adjoining lands shall only be done if granted previous permission on the private side of the boundary.

Boundary Maintenance Levels:

1. **Level 1 - Difficult:** 0 - 75% of previous posting is in place or original position is evident, or previous posting may have been moved and survey equipment is required to maintain. Extensive brushing of line needed. See [.860 Boundary Line Resurvey](#).
2. **Level 2 - Normal:** 75 - 95% of previous posting and blazing is in place or original position is evident and few gaps between existing post greater than 600 feet (see [.850 Gaps in Sign Posting](#)), with moderate brushing of line needed.
3. **Level 3 - Easy:** 95% or more of previous posting and blazing is in place with minimal brushing of line needed.

.811 Clearing

Lines shall be cleared in accordance with [.523 Clearing](#).

.812 Blazes

Blazes shall be added to new trees to bring the line into compliance with [Exhibit .522](#) and the spacing requirements of [.522 Blazes](#). Previously blazed trees shall be preserved and not be re-blazed if the old marks

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are readily discernable. New blazes shall be added only where old marks are not readily discernable. Blazing on the adjoiner's side of the line shall not be performed if the adjoiner denies permission to mark their side of the line.

.813 Painting

Painting shall be in accordance with [.524 Painting](#). Existing paint (not applied under this contract) found in the correct location shall be repainted after all the loose material is scraped away (can use the bottom of paint to scrape the loose material away prior to painting). Existing paint on public land and all paint known to have been placed by Government procured surveys (on either public or adjoining lands) found out of compliance with [.524 Painting](#) and [Exhibit .522](#) shall be covered with black paint by the Contractor.

.820 Boundary Posts and Signs

Maintenance of previously posted lines includes rehabilitating the posts, so they are plumb and firmly set and replacing property boundary signs which have become unreadable, unsightly, or otherwise unserviceable. All unserviceable boundary marking materials shall be removed from the project area and returned to the COR.

Signs to be used are shown in [Exhibit .423](#). All existing sign posts maintained, and new sign posts added shall comply with [.530 Posting the Property Line](#) and [Exhibit .522](#), including guard posts. Pink flagging shall be attached to the top of all sign posts, existing or new, located on the boundary line.

.821 Existing Sign Posts:

Existing sign posts found along the line shall be at a minimum, visually evaluated to determine whether they are within 1 foot of the boundary line, in accordance with [.531 Post and Sign Placement](#). If so, they shall be straightened, driven, and/or firmed up. When sign posts are found disturbed or displaced, but otherwise undamaged, the Contractor shall re- drive the post within 1 foot of the boundary line.

Older boundary line sign posts may have been set under less restrictive positional standards, such as the sign posts only being required to be within 2 feet of the boundary line, or the maximum spacing being no more than 300 feet. The COR shall notify the Contractor of lines suspected to have been signed and posted under less restrictive standards. The Contractor may, at the discretion of the Government, be requested to maintain these lines in accordance with [TECHNICAL SPECIFICATIONS - MAINTENANCE](#), or to re-survey and re-post these lines in accordance with [.860 Boundary Line Resurvey](#). An equitable adjustment of contract performance time or price may be made for any addition or deletion of survey work authorized by the CO resulting from boundary lines being previously posted under less restrictive standards.

.822 New Sign Posts:

When their original position can be determined, sign posts that are in damaged or deteriorated condition shall be replaced by a new sign post. When the original position of sign posts that are in damaged or deteriorated condition cannot be determined a new sign post shall be driven in accordance with [.531 Post and Sign Placement](#). Damaged and deteriorated signs and sign posts shall remain the property of the Government and shall be returned to the COR, unless the COR directs otherwise.

If the sign posts found in place are not durable in nature such as deteriorating fiberglass or substandard flimsy posts; a new sign post shall be placed on-line near the old post. If T-Bar posts are found, it is acceptable to attach/reattach property boundary signs using steel wire.

.823 Sign Attachment:

The Contractor shall ensure that existing signs on sign posts are securely attached and in good readable condition. Signs that are damaged or deteriorated shall be replaced with the appropriate sign. The Contractor

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shall firmly attach signs to sign posts with corrosion resistant bolts and nuts, aluminum rivets, or corrosion resistant wire as directed by the COR.

.830 Erroneously Located Sign Posts

The Contractor shall notify the COR of any existing sign posts previously placed by survey found to be more than 1 foot off the boundary line, report the distance they are off-line, and provide an opinion as to the likely cause, such as innocent error, moved by landowner, trespass in vicinity, etc. The COR will specify what corrective actions the Contractor shall take in any such cases. In the absence of direction from the COR the Contractor shall move the sign posts to the boundary line and drive them in the ground in accordance with [.531 Post and Sign Placement](#).

Said sign posts which may have encouraged either encroachment or prohibited activities within a wilderness area shall be reported to the COR as per [.130 Conflicts, Boundary and Title Disputes](#).

Refer to [.540 Obliteration of Previously Posted Lines](#) for signs not known to be previously placed by survey.

.840 Encroachments

All encroachments/trespasses found or suspected shall be described, diagramed, and photographed. The approximate location shall be reported in latitude and longitude using decimal degrees (NAD 83) to an accuracy of 5 decimal places.

.850 Gaps in Sign Posting

Where there is a gap of 200 to 600 feet between existing sign posts which are found to be within 1 foot of the boundary line the Contractor shall clear, blaze, paint, and place posts in the gap until the line is brought into compliance with [.530 Posting the Property Line](#) and [Exhibit .522](#). The placement of any sign posts which requires the measurement of angles, distances, or positions, shall be performed by or under the direct supervision of a licensed land surveyor.

The Contractor shall not blaze or place posts in gaps greater than 600 feet and shall notify the COR of the gap(s) as soon as is practical. After notification the Contractor may, at the discretion of the Government, be requested to re-survey and re-post the boundary line in the gap(s). Re-survey and re-posting shall be performed in accordance with [.860 Boundary Line Resurvey](#). In such cases an equitable adjustment of contract time and/or price will be made for the addition of the survey work authorized by the CO.

.860 Boundary Line Resurvey

The Contractor shall perform work on all lines identified for boundary line re-survey on Exhibit 3 in accordance with [.300 Control survey, field data collection and computations](#) and [.500 Boundary Line Marking and Posting](#). The work shall be performed by or under the direct supervision of a licensed land surveyor.

Boundary lines are to be re-surveyed from monument to monument, unless directed otherwise by the COR. When required by State or local statutes, or by the COR, the Contractor shall prepare and file a plat for the lines resurveyed in accordance with [.600 Plats](#).

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TECHNICAL SPECIFICATIONS – MISCELLANEOUS

.900 MISCELLANEOUS SERVICE

.910 Surveyor's Report

The Contractor shall furnish the Forest Service a Surveyor's Report similar to that shown in [Exhibit .910](#). The report shall contain any information pertinent to the project. It is not necessary to repeat the detailed information contained in other records created and filed as part of this contract. At a minimum, the report shall contain the following:

1. The information contained on [Exhibit .910](#).
2. Results of search for all corners.
3. Visible signs of encroachment or trespass.
4. Statements relevant to the project made by adjoining landowners or other interested parties.

.920 Encroachment Diagram

An encroachment/trespass is any apparent structures, significant improvements, or unauthorized occupancy on National Forest System Lands. An encroachment diagram is required in pertinent areas to show in greater detail the difference between lines of occupation and lines of title. All features that indicate occupation where it conflicts with title shall be shown with a stationing scheme or leaders showing posts set and the location of potential conflicts, encroachments, trespass, or unauthorized use in relation to the nearest survey corners as shown on [Exhibit .130](#).

Additionally, the approximate location of the encroachments shall be reported in latitude and longitude using decimal degrees (NAD 83) to an accuracy of 5 decimal places. Photos will be provided to the COR as per [.930 Photos](#).

.930 Photos

The Contractor shall take digital photos of all new monuments and accessories, potential conflicts, encroachments/trespasses, and any other items of interest the Contractor deems necessary. The photos shall be cataloged or named in a manner where the location or reason for the photo can be readily known, and shall be supplied to the COR by agreed upon media (i.e. CD, flash drive, internet drive, email).

The COR may request the Contractor to furnish digital pictures of any conflict as described in section [.130 Conflicts, Boundary and Title Disputes](#) and all physical evidence found at corners and their accessories identified on Exhibits 2 and 3. An object of recognizable scale will be in pictures when identifying size of an object is of primary concern.

.940 Project Documentation

All relevant data to this survey will be furnished to the Forest Service in a digital format (USB drive or internet based shared folder) that includes Corner Recordations, Certificate of Surveys, Digital Photographs of Corner Remonumentation, Survey Reports, Calculations, Control Diagram, Final Latitude and Longitude for all corners tied and set in the survey.

.950 Quality Control, Inspection & Acceptance

.951 Quality Control

The Contractor has sole responsibility for maintaining quality control for all aspects of work performed under this contract. The Contractor will furnish a written plan of self-inspection procedures to the COR on request.

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The COR or designated inspector will inspect the work for compliance to answer a yes/no question as to if the work is within the contracted specifications, not to serve as quality control.

.952 Inspections

The COR or a designated Inspector will make periodic inspections of each project as work progresses. The Contractor shall give the COR ample time to schedule inspections for compliance prior to submitting any invoices.

The Contractor shall use professional judgment in determining the position of any corner to be established or re-established under this contract in accordance with the BLM Manuals and State rules and regulations. If under subsequent review by the COR this judgment is not agreed to, then the situation, e.g., move corner monument and/or line posts, shall be rectified at the Contractor's expense.

.953 Acceptance

No final approval of any project will be granted until all items specified are delivered to the COR and determined to fulfill all specifications, terms, and conditions of this contract. Acceptance is required prior to payments; the Contractor will confirm acceptance from the CO/COR prior to submitting invoices. During larger or long-term contracts, intermediary portions of acceptance might be requested for payment purposes, the amount of the portion will be agreed upon with the CO/COR.

All other surveys provided by the Contractor shall be in accordance with applicable State Minimum Standards of Practice providing they are not less than requirements set forth in these specifications or subsequent Supplemental Specifications or Special Instructions and exhibits.

.960 Other Surveys

Engineering surveys include, but are not limited to, the following: topographic surveys, right-of-way surveys, photogrammetric control surveys, bathymetric surveys, etc. Supplemental Specifications or Special Instructions will be issued for these and other surveys as necessary.

.961 Topographic Surveys

1. **Survey Limits:** Survey limits will be furnished to the Contractor on previous map or sketch of project site, and/or flagged on the ground by the Government.
 - a. Prior to actual field work, the Contractor will consult with the COR to determine the scope and details of the project. This will ensure that the final product conforms to the expectations of the COR.
2. **Survey Control:** A Primary Control point is selected in or near the project area. This point's absolute location shall be determined to within 0.25 feet horizontal and 0.50 feet vertical precision at the 95% confidence level, by OPUS or other suitable means.
 - a. COR may waive requirement for Primary Control for Topographic Surveys. In this case Contractor will assign "assumed" coordinate values at the initial minor control point. Coordinates shall be selected which allow the entire project to be mapped without entering negative coordinate values. Elevation assigned should not approximate mean sea level as on a USGS topographic map.
 - b. Minor control points shall be established inter-visibly as needed to accomplish the Total Station survey. If the topographic survey may be accomplished completely by GPS, Contractor shall nevertheless set sufficient control points to allow future construction Contractor's surveyor to place the design on the ground. At a minimum, this requires 2 inter-visible points for a small site, or 2 sets of 2 inter-visible points (2 Azimuth pairs) at opposite ends of a more linear project.

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True ground distances shall be measured such that resulting mapping may be generated at a project scale of 1.0 exactly.

- c. Control points shall be set with a durable marker, such as rebar with a metal cap, or large nail with a reference point washer, detectable with a magnetic locator. The marker cap or washer should be stamped with a unique identifier which is the point name or number, such as "CP 1".

3. *Features to Survey:*

- a. Sufficient topo points to generate a topographic map of the project at a contour interval of 1 foot, or another interval as specified by the COR.
- b. Generally, all artificial features will be located with basic description including, but not limited to:
 - i. Roads and trails, along with pavement type. Unimproved trails to be located along centerline only and provide typical width.
 - ii. Buildings or other structures, including toilets and sheds; main footprint and type, e.g. toilet, picnic shelter, etc.
 - iii. Bridges with width and type of construction.
 - iv. Culverts and other drainage or irrigation structures with size and type, e.g. 12"CMP, concrete weir, etc.
 - v. Water utility features, e.g. well, hand pump, water valve, etc.
 - vi. All utility poles, markers, fixtures; state overhead or buried and type of utility if known or evident e.g. electric, telephone, gas, etc.
 - vii. Concrete or paved pads of any sort; state purpose, e.g. dumpster pad.
 - viii. Recreational use features such as picnic tables, tent pads, playground fixtures.
 - ix. Fences, including type of construction.
 - x. Signs, including traffic, interpretive and bulletin boards. Multiple panel signs require several shots. Note traffic sign wording. Can be presented on map in tabular form.
- c. Individual trees in campgrounds may or may not need to be located; check with COR. Where site is partly trees and partly open, collect tree lines as a feature string and note types.
- d. Natural features including rock outcrops, large boulders, streams and ponds, marshes, and within recreation or residential sites, areas of standing water or poor drainage.
- e. Nearest utility: where site is not currently served by a utility but a nearby connection or marker is visible from site control, such feature shall be located and described, but need not be included within the limits of the contours or terrain model. Features which fall outside map sheet extents may be described by coordinates or bearing & distance.

4. *Break Lines and Feature Strings:*

- a. Break Lines are lines where the ground surface deflects, or changes grade, and are variously referred to as topo breaks or fault lines. Examples of Break Lines are the toe of a slope, the top of river bank, and the flow line of a gutter.
- b. Feature Strings are lines which define or bound a physical feature or object and are also referred to as polylines or figures. These planimetric features are lines that should be drawn on the topographic map, usually manmade features, but could also be the edge of a water body. Examples of Feature Strings are edge of pavement, dirt trail, fence line, building line. All Break Lines and Feature Strings encountered in the survey shall be collected in such manner, and delivered electronically to the Government such that Government designers can incorporate these lines in their design software.

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5. *Topographic Map Deliverables:*

- a. Paper copies of the map, in color, shall be provided, as requested by the COR.
- b. The map shall also be furnished in .DWG file format. This file shall be provided on USB drive or other means specified by the COR.
- c. Map features should be separated into thematic layers generally based on surface materials or function with simple common language names such as "contours", "break lines", "trees", "concrete", "utilities", etc.
- d. Specific CAD standards are not required by the Government, but final product map colors and symbology should generally mimic USGS topo maps.
- e. A table of Control points with coordinates and descriptions shall appear on the map, and also be supplied digitally in text format, including a line of column or field headings.

.962 Right-of-Way Surveys

Rights-of-Way surveys shall be surveyed along existing or proposed centerlines, and shall be platted and described in terms of tangent bearings, distances, deflections, and/or simple curves parallel to the surveyed line, and shall not be monumented unless otherwise directed by the CO. All Rights of Way surveys shall be tied to the appropriate property corners of the subject properties.

1. The surveyed centerline shall not exceed 3 feet in any location from the actual centerline location of the existing road or trail or the flagged proposed road or trail location.
2. A separate plat or exhibit shall show the Right-Of-Way (ROW) survey. The plat shall give the total ROW area, pertinent curve data and the other measurements, ties to controlling PLSS, Mineral Survey, HES, or other corners, and conform to existing applicable State Statutes. The width of the ROW shall be supplied by the task order or COR. See [Exhibit .962](#) for ROW survey checklist.
3. Any encroachments found within the ROW shall be noted on the ROW plat or exhibit.
4. A Legal Description shall be prepared in the form approved by the COR. The description will include ties from the Beginning and End points to nearby PLSS corners or other appropriate corners.
5. Any deviations from this requirement must be approved by the COR or shall be addressed in the Task Order Scope of Work if changes are requested by the Government. GPS technology may be used as determined by the COR.

.963 Photogrammetric Control Surveys

1. ***Survey Types and Definitions:*** Surveys fall into one of two stages; the aerial photography, and the ground control survey. The sequence will vary due to seasonal constraints, and the schedule constraints of both the aerial and ground survey providers. The actual scope of the project may vary and shall be determined by the COR.
 - a. Prior to actual field work, the Contractor will consult with the COR to determine the scope and details of the project. This will ensure that the final product conforms to the expectations of the COR.
2. ***Control Survey:***
 - a. A Primary Control point is selected in or near the project area. This is typically used as a GPS base point for RTK or Fast-Static survey and may or may not be one of the required photo control points. This point's absolute location shall be determined to within 0.25 feet horizontal and 0.50 feet vertical precision at the 95% confidence level, by OPUS or other suitable means.
 - b. Panels or photo "id points shall be located relative to said Primary Control to within 0.2 feet horizontal and 0.3 feet vertical precision at the 95% confidence level. True ground distances shall be measured such that resulting mapping may be generated at a project scale of 1.0 exactly.

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3. **Required Locations:** The COR will provide Contractor with a map of required control point locations and required target type and size. If targets are placed by the Government, recreation-grade GPS coordinates will be provided to facilitate finding the targets. Contractor has some latitude, typically up to 200' depending on photo scale, as to actual target location as long as the panel falls within the photo limits. The Contractor shall select such location so as to allow target to lie as flat as possible, be clearly visible from the sky and clear of overhanging trees and their shadows. For photo-id projects, prints of the actual project photography will also be provided.
4. **Physical Requirements:**
- a. Contractor shall supply all target materials.
 - b. Target overall size will be provided by COR.
 - c. At the center of each target shall be set a durable marker, such as rebar with a metal cap, driven flush with the panel surface, except where ground conditions do not allow such marker.
 - d. The marker cap or washer should be stamped with a unique identifier which is the point name or number, such as "CPI".
 - e. Contractor shall note any variation between the target elevation and the control point monument in excess of 0.1 feet; also note any significant elevation disparity across the panel.

5. **Removal of Panels:** Targets are to be removed following the photo flight.

.970 Special Designated Area Boundary Surveys

The survey, monumentation and posting of Special Designated Areas such as Wilderness, Wild and Scenic Rivers, Research Natural Areas, etc., their Boundaries shall be conducted using technical specifications that, in part, depend on the vicinity to adjacent property rights, topography, and calls in the legal description. In addition, the following shall apply:

.971 Vertical positioning

The method used to establish corners or boundary lines at or along a specified elevation contour shall be determined by the Contractor and shall meet the criteria of plus or minus 3 feet (unless otherwise directed by the COR) from elevation called for in the legal description. Additional specifications may be directed by the task order, especially in areas of flat topography.

.972 Monument Caps

Monument caps and tablets shall be stamped as directed by the COR. Each wilderness boundary cap shall be stamped with the CDA wilderness name, angle point number, date established and the PLS number of the surveyor establishing said monument. Monuments shall be placed at the locations shown on Exhibit 3.

.973 Signing

Signs to be used are shown in [Exhibit .424](#). The appropriate sign designating the area (e.g. "National Forest Wilderness") shall be attached to trees or posts as directed by the COR. Signs shall be placed at ridge, draw, road, and trail crossings. Signs shall be spaced a maximum of 200 feet apart or as directed by the COR and placed to be inter-visible if brush and trees were removed.

*Intermountain Region 4
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SECTION J - EXHIBITS

J-2 TABLE OF EXHIBITS FOR CADASTRAL SURVEYS

***EXHIBITS TO BE CONTAINED IN THE TASK ORDER PACKAGE**

EXHIBIT 1	*VICINITY MAP
EXHIBIT 2	*CORNER AND OWNERSHIP STATUS
EXHIBIT 3	*WORK TO BE ACCOMPLISHED

EXHIBITS CONTAINED IN THIS DOCUMENT

<u>EXHIBIT .120</u>	<u>EXAMPLE LETTER TO ADJOINERS & REPLY CARD</u>
<u>EXHIBIT .130</u>	<u>EXAMPLE CONFLICT REPORT/TRESPASS DIAGRAM</u>
<u>EXHIBIT .220</u>	<u>EXAMPLE FS CORNER CARD (Form 7100-52)</u>
<u>EXHIBIT .411</u>	<u>PLACEMENT OF CORNER MONUMENTS</u>
<u>EXHIBIT .412</u>	<u>TYPICAL PLACEMENT OF REFERENCE MONUMENTS</u>
<u>EXHIBIT .423</u>	<u>BEARING TREES</u>
<u>EXHIBIT .424</u>	<u>SIGNS</u>
<u>EXHIBIT .430</u>	<u>EXAMPLE IDAHO CORNER FORM</u>
<u>EXHIBIT .521</u>	<u>SIGN PLACEMENT</u>
<u>EXHIBIT .522</u>	<u>MARKING AND POSTING THE BOUNDARY LINE</u>
<u>EXHIBIT .600</u>	<u>EXAMPLE PLAT</u>
<u>EXHIBIT .620</u>	<u>CADASTRAL PLAT CHECKLIST</u>
<u>EXHIBIT .910</u>	<u>EXAMPLE SURVEYOR'S REPORT</u>
<u>EXHIBIT .962</u>	<u>RIGHT-OF-WAY SURVEY CHECKLIST</u>

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EXHIBIT .120 - EXAMPLE LETTER TO ADJOINERS & REPLY CARD

Example of Letter to Adjoiners, page 1 of 2

(Date)

(Landowner)

1234 XXXXXX Road

City, State Zip

RE: XXXX Land Survey Project

Township XX South, Range XX East, W. M., Section XX, Tax Lot XXXXXXXXXXXX

Dear (Landowner):

Over the next year(?), the USDA Forest Service will be conducting boundary survey work in the area adjoining your property. We expect this work to begin by (date).

Our firm has been contracted by the USDA Forest Service to retrace the boundary lines of the XXXX National Forest in the area adjoining your property. We will be working in this area during the month of (MONTH).

If you have any information concerning the boundary line that you would like us to be aware of, or any desires you wish to express please contact our firm to arrange a date and time for a meeting.

Surveying and marking National Forest boundaries include the following standards:

... Clear underbrush 2 feet on either side of the line,

... Blazes on trees within 3 feet of the line.

... Boundary line signs within 1 foot of the line.

... Set new Forest Service monuments and mark new bearing trees where applicable.

We ask your permission to perform this work on your side of the boundary line. Please fill out the enclosed card, either granting or denying us permission to mark your side of the boundary line, and return the card to our office in the enclosed addressed envelope.

Thank you in advance for your cooperation.

Sincerely,

XXXXXX

/s/ XXXXXXXXX, PLS

Land Surveyor/s/

*Intermountain Region 4
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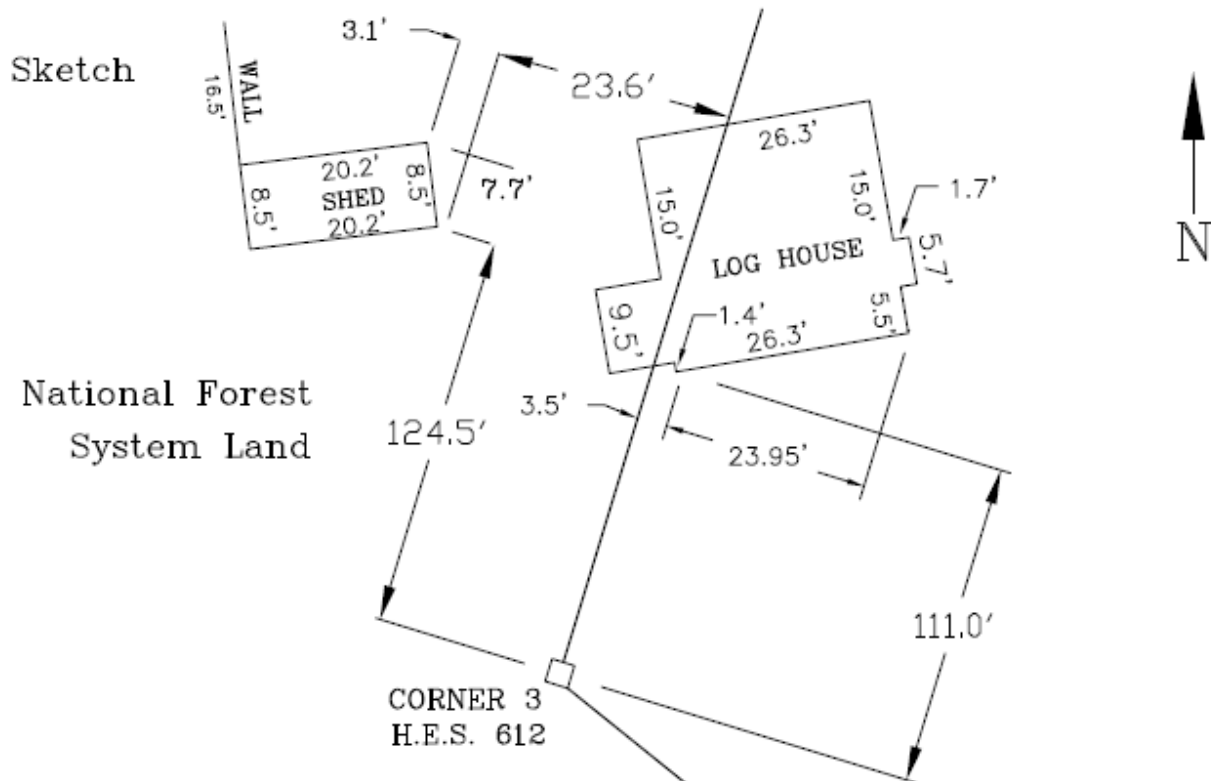
EXHIBIT .130 - EXAMPLE CONFLICT REPORT/TRESPASS DIAGRAM

Sample Surveys, Inc.

Registered Land Surveyors

Identification of Possible Conflict

Contract Number: <u>53-67T0-6-05</u>	Date Discovered: <u>11/11/2003</u>
Location: <u>Vicinity of Corner 3 of H.E.S. 612, Section 14 T5S, R45E PMM</u>	
Name\Location of Adjoining Landowner: <u>C.D. Green</u>	
<u>320 Ryan Road</u>	
<u>Ashland, MT 12345</u>	
Nature of Conflict: <u>Structures located on National Forest Land.</u>	
Remarks: _____	



Submitted By: _____
Received By: _____

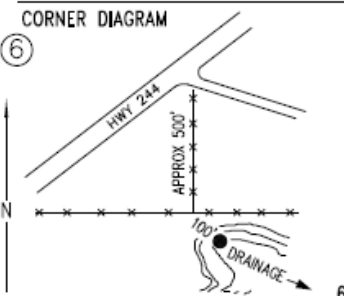
Date: _____
Date: _____

Intermountain Region 4
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EXHIBIT .220 - EXAMPLE FS CORNER CARD (Form 7100-52)

Front - FS Corner Card, page 1 of 2

FRONT

USDA-FOREST SERVICE		CORNER RECORD (SEARCH) (Ref. FSM 7151.4)		Tract or Survey No. _____	Cor. No. <u>1</u> ①
Index File No. <u>C-18</u> (sections only)		R.D. <u>PACTOLA</u>	County <u>PENNINGTON</u>	ALSO CW 1/16 SECTION 2	
T. <u>0S</u> <u>0E</u> R. _____		N.F. <u>BLACK HILLS</u>	State <u>SOUTH DAKOTA</u>	Declination <u>12° East</u> 1	
Meridian <u>BLACK HILLS</u>					
⑥ CORNER DIAGRAM 		DESCRIBE CORNER MONUMENT FOUND ② FOUND THE BASE OF THE ORIGINAL LIMESTONE 12" x 6" x 8", FIRMLY SET, FLUSH WITH GROUND, OVER A BROKEN POTTERY MEMORIAL. TOP SECTION WITH SCRIBING GONE. (description of monument found: type, material, size, how set (loose or firm), above ground dimensions, how marked or scribed, any other related facts)			
⑦ PHOTO IDENTIFICATION DATA Photo No. _____ Filed At. _____ Object Ident. _____ Ident. by <u>Signature</u>		DESCRIBE BEARING TREES OR OTHER OFFICIAL ACCESSORIES FOUND ③ A SCATTERED MOUND OF STONE BEARS SW, 4 FT. A 16" PINE; SCRIBING HEALED OVER; BEARS N 63 E, 18.5 FT. MATCHES RECORD A 10" PINE SNAG, 5 FT. TALL; SCRIBED "HES 000 BT"; BEARS S 60 E, 65 FT. RECORD CALL IS N 60 W, 65 FT.			
⑧ PLANE COORDINATE POSITION DATA State _____ Zone _____ X _____ Y _____ Estab. By _____ <u>Agency</u>		DESCRIBE CORNER LOCATION RELATIVE TO NEAR-BY FEATURES, ALSO HOW TO REACH CORNER ④ FROM CARSON DRIVEWAY INTERSECTION WITH HWY 244, FOLLOW FENCE LINE SOUTH ALONG FOREST EDGE TO THREE WAY FENCE CORNER, APPROX 500 FT. THE CORNER SETS 100 FT SE IN THE BOTTOM OF A SMALL DRAINAGE, RUNS SE, IN SCATTERED PINES.			
⑤ Signature _____		Title R.L.S.	Division AAA SURVEYS	Headquarters PRINGLE SD	Date 10-9-91
FS-7100-52 (3/68) 5					

INSTRUCTIONS—Numbers refer to those indicated on the card.

1. Self-explanatory. R.D. denotes Ranger District and N.F. denotes National Forest.
2. Describe the monument found.
3. Describe the accessories found.
4. Describe general location, how to reach, local witnesses and general information.
5. The person or contractor, and date of recovery.
6. Sketch showing major topographic features and B.T.'s in approximate relationship to corner.
7. Identify aerial photograph and tie to picture point. Required when corner record card (7100-52) and photo identification of corners are both required by the contract.
8. Self-explanatory. Required only if the contractor uses state plane coordinates in survey computations.

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EXHIBIT .220 - EXAMPLE FS CORNER CARD (Form 7100-52)

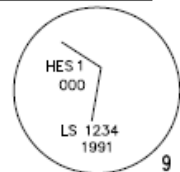
Back - FS Corner Card, page 2 of 2

BACK

CORNER RECORD (PERPETUATION)

⑨ MONUMENT: DESCRIBE NEW MONUMENT SET, OR WORK DONE TO PRESERVE EXISTING MONUMENT

SET A STANDARD USFS ALUMINUM MONUMENT, 30" LONG, 22" IN THE GROUND, OVER A "DEEP ONE" MAGNETIC MEMORIAL. BURIED ORIGINAL STONE AND POTTERY ALONGSIDE.
USED OLD STONE MOUND TO BUILD A 3' FT DIA. COLLAR AROUND MONUMENT. CAP MARKED:
SET LAND SURVEY MONUMENT AND BOUNDARY SIGN ON RED STEEL POST 1 FT NE.



⑩ ACCESSORIES: DESCRIBE NEW BT's WITNESS OBJECTS ETC. ESTABLISHED, OR WORK DONE TO PRESERVE EXISTING EVIDENCE

A 6" PINE; SCRIBED "I HES 000 BT"; BEARS N 46 W, 10.2 FT.
FOUND A 5/8" REBAR; 6" ABOVE GROUND; BEARS N 47 W, 98.2 FT. SEE REMARKS
PUT TAGS ON BT's AND SNAG.

10

⑪ Work Done By J. SURVEYOR Title SDRLS Date 9-8-91 State Registration No. 1234
or Agency 1234
Certified Cor. Record Prepared? ☒ Yes ☐ No Filed At: PENNINGTON COUNTY COURTHOUSE Date 10-91

11

⑫ REMARKS

REBAR NOTED WAS SET BY LS 66 IN 1965 DURING A PRIVATE SURVEY OF HES-000. THE PLAT ON FILE INDICATES THIS CORNER WAS SET BY IMPROPER METHODS. HAVING FOUND AND CONFIRMED THE ORIGINAL MONUMENT, I REJECT THIS REBAR.

12

⑬ Index File No. C-18 T. 0S R. 0E M. B.H. Tract of Survey No. HES 000 Cor. 1

13

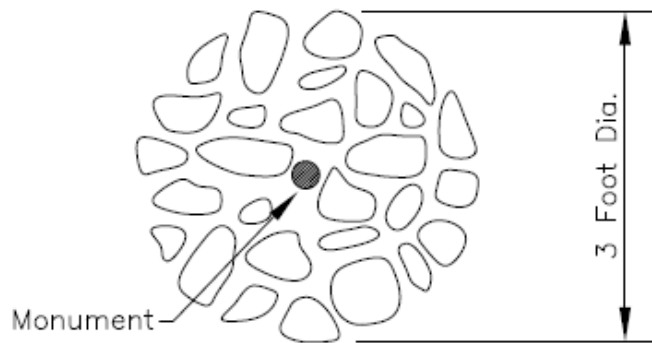
INSTRUCTIONS—Numbers refer to those indicated on the card.

9. Describe work done in remonumenting, refurbishing, or remarking the corner. Fully describe any new monument set.
10. Describe work done to preserve old bearing trees, and fully describe all new bearing trees and objects set.
11. The registered surveyor who accomplished the work, or who supervised it.
12. May be used for information from front of card or other comments.
13. Self-explanatory.

Note: Cards may be typed or completed neatly in ink.

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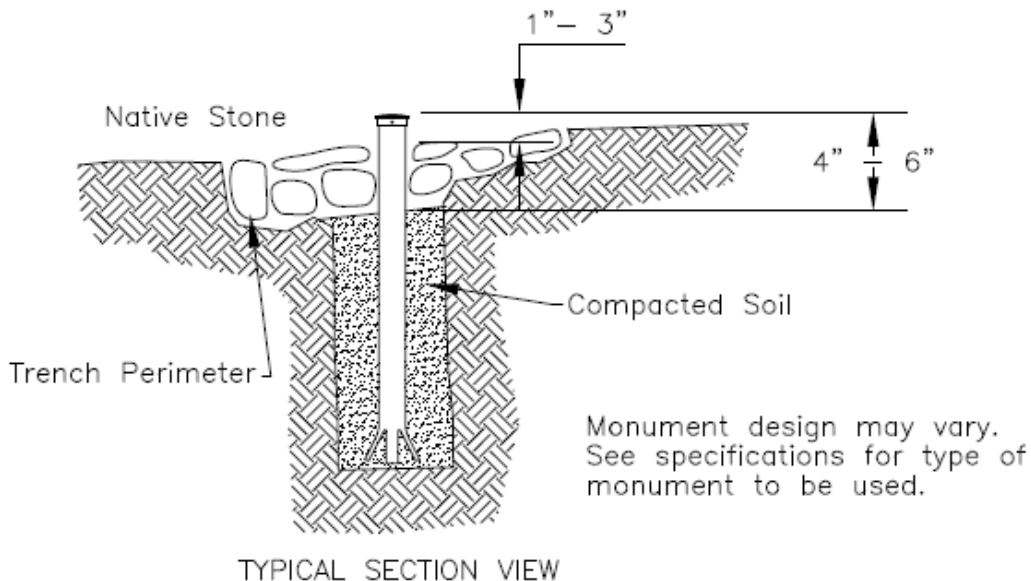
EXHIBIT .411 - PLACEMENT OF CORNER MONUMENTS



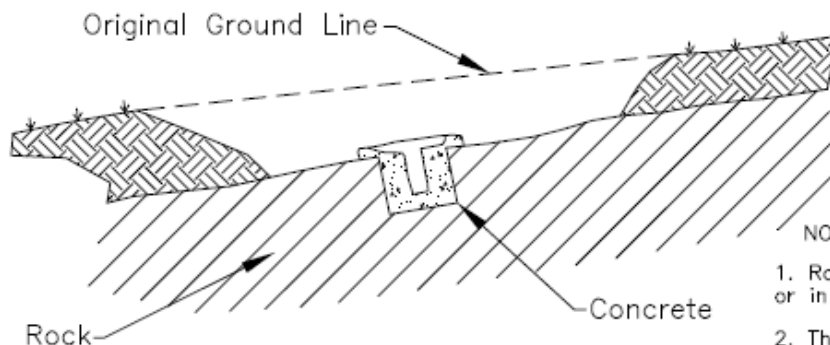
Monument w/Mound of Stone – Typical
PLAN VIEW

NOTES:

1. Native stones within 100' of the corner shall be used to construct a mound around the monument. Where the number of available stones is insufficient to construct a 3 foot diameter mound, the contractor shall construct a mound as large as the available stones will permit.
2. Monuments shall be plumb with the backfill soil compacted to 4" to 6" layers.



TYPICAL SECTION VIEW



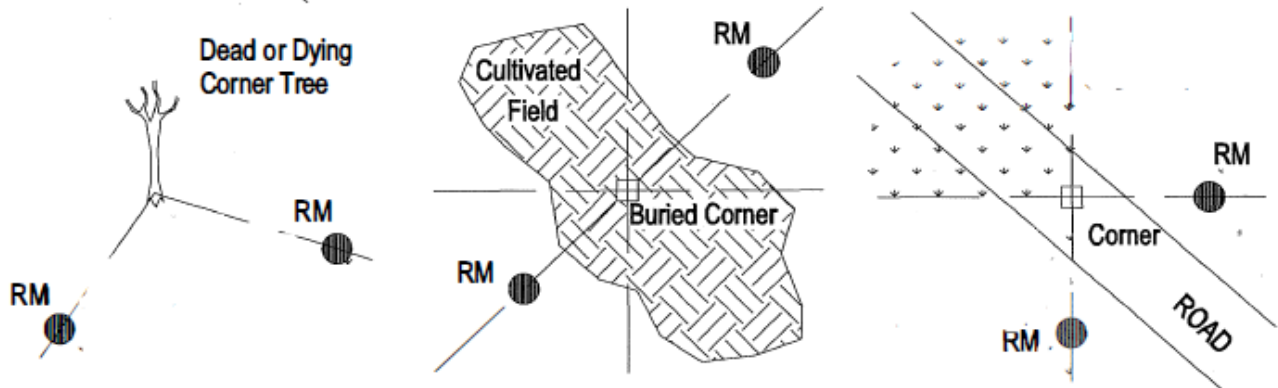
TYPICAL ROCK TABLET INSTALLATION

NOTES:

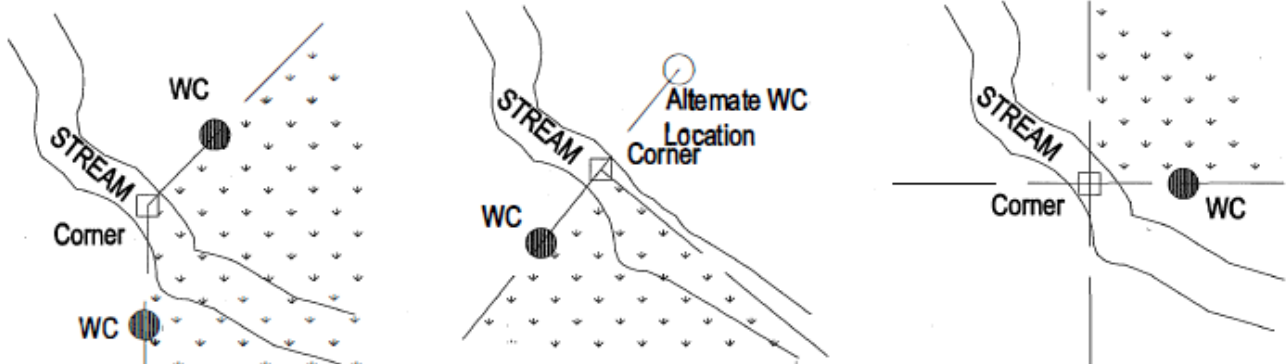
1. Rock tablets shall be used in solid rock or in place stones impractical to remove.
2. The contractor shall clean the rock surface and provide for drainage.
3. Rock tablets shall be flush to surface.

EXHIBIT .412 - TYPICAL PLACEMENT OF REFERENCE MONUMENTS

TYPICAL PLACEMENT OF REFERENCE MONUMENTS

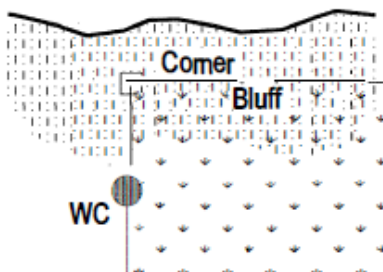


TYPICAL PLACEMENT OF WITNESS CORNERS



NOTES:

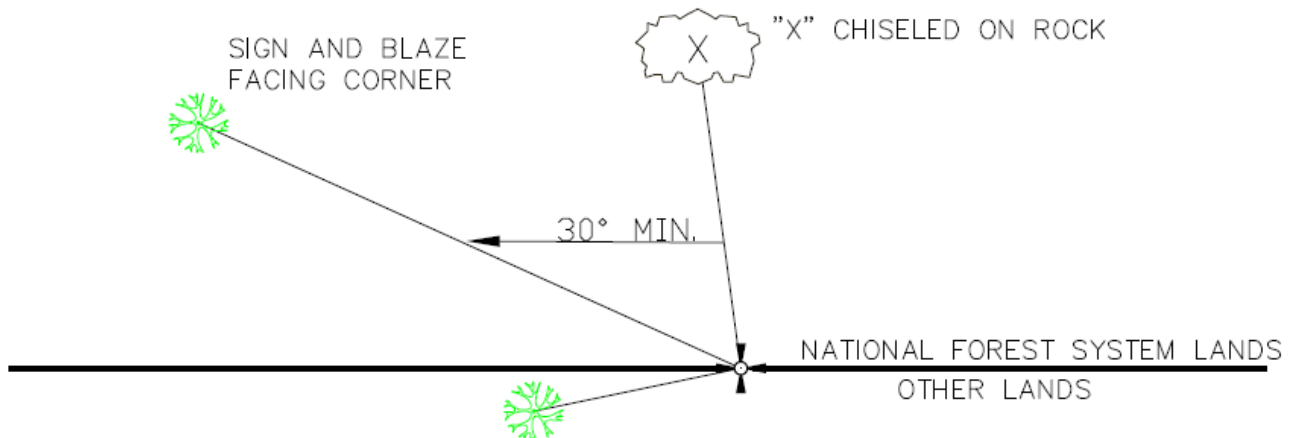
1. In sectionalized land the Manual shall be used.
2. Witness corners shall be placed on the Forest Service property line if practical. The COR may direct they be set in an alternate location.
3. Reference monuments shall not be placed on or near the Forest Service property line.
4. A guard post with sign 54-9, Land Survey Monument, shall be placed at all reference monuments and witness corners.



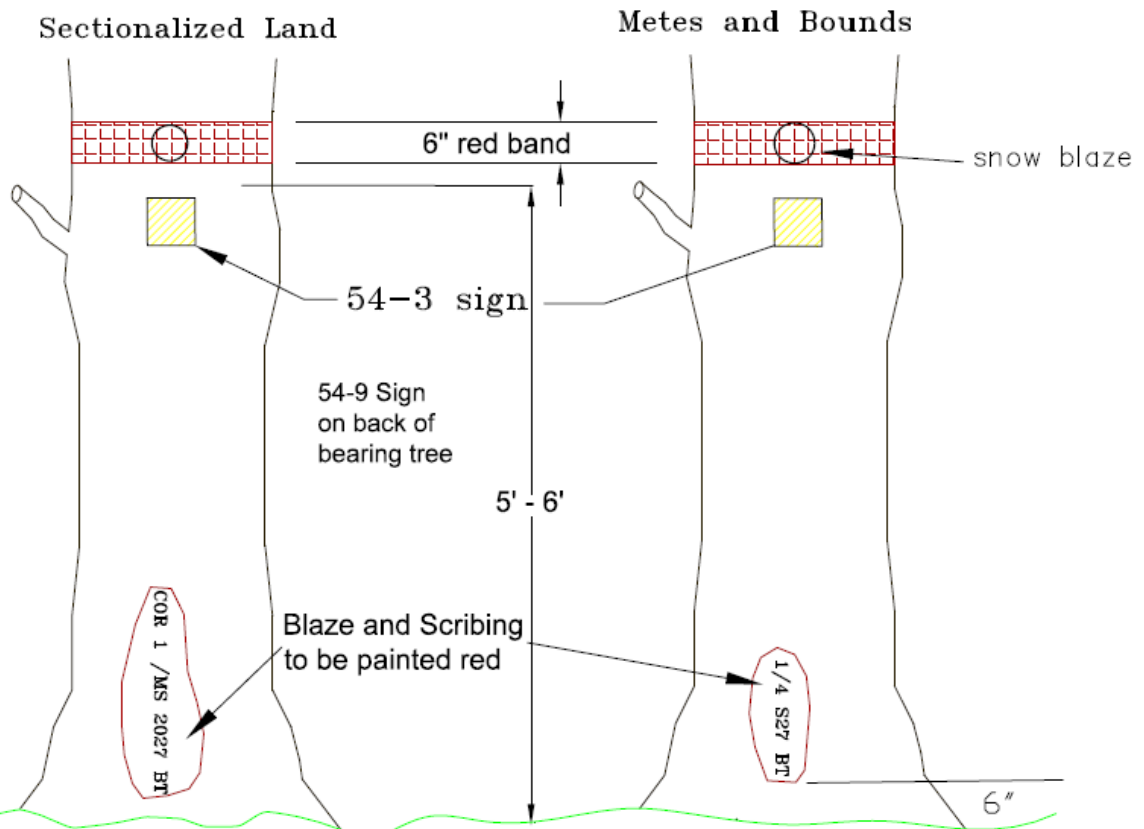
National Forest Land

*Intermountain Region 4
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EXHIBIT .423 - BEARING TREES



TYPICAL BEARING TREES ON NATIONAL FOREST SYSTEM LANDS



Notes:

1. See Manual for Specific markings.
2. Smooth bark Maple, Alder, Aspen, etc..., shall not be blazed, but shall be scribed and painted as if blazed (bark scribed).
3. Original bearing trees shall not be rescribed.
4. All bearing trees are to be painted with a 6" red band. The band shall be located around the tree approximately 6' above the ground. Rough bark trees shall have the area of the band smoothed before painting.
5. Red paint shall be neatly applied over blazes and scribing.
6. The contractor shall stamp or etch the required information on the 54-3 sign.
7. Signs shall be attached to bearing trees with 2 Aluminum nails per sign. Leave nails protruding at least 1/2" on trees to allow for growth.

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EXHIBIT .424 - SIGNS



54-2



54-7



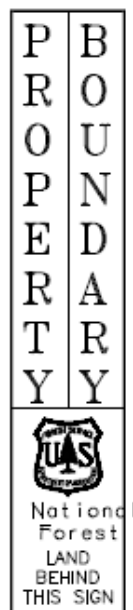
54-9



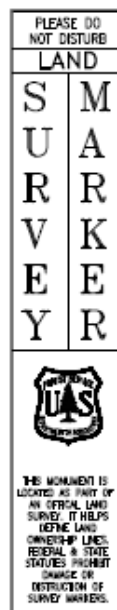
P54-3
(DECAL)



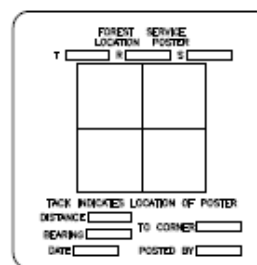
P54-2
(DECAL)



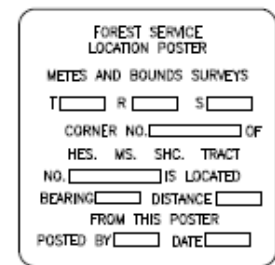
P54-1
(DECAL)



P71-10
(DECAL)



54-6



54-8



27-6A



54-3



54-5

Intermountain Region 4
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EXHIBIT .430 - EXAMPLE IDAHO CORNER FORM

STATE OF IDAHO
CORNER PERPETUATION AND FILING RECORD

(In compliance with Title 55, Chapter 16, Sections 1601-1612 CORNER PERPETUATION AND FILING ACT.)

RECORD OF ORIGINAL CORNER: Date of Survey 1970 Surveyor BLM

Corrective Dependent Resurvey by BLM in 1987.

0.00 Chs. The witness cor. of secs. 27 and 28, monumented with an iron post, 2 1/2 ins. diam., firmly set in the ground and in a mound of stone, 4 ft. base, to top, with brass cap, from which 1970 bearing trees

A pine, 9 ins. diam., bears S.20°W., 80 lks. dist., with a partially healed face.

A fir, 24 ins. diam., bears N.75°W., 33 lks. dist., mkd. X BT.

2.10 Chs. True point for the 1/4 sec. cor. of secs. 27 and 28, determined on a line projected through the witness cor., at the 1970 record. This point falls in Lake Pend Oreille where it is impracticable to establish a permanent monument.

DESCRIPTION OF CORNER EVIDENCE FOUND:

A 2 1/2" diam. iron pipe with a 3" diam. brass cap, in a mound of stone, from which

A pine snag, 14 ins. diam., at record bearing, 44.2 ft. dist., with a healed face.

A Douglas fir, 24 ins. diam., bears N.57°W., at record dist., marked BT.

DESCRIPTION OF MONUMENT AND ACCESSORIES ESTABLISHED TO PERPETUATE THE LOCATION OF THIS CORNER.

Two new bearing trees:

A Douglas fir, 27 ins. diam., bears S.86°W., 16.5 ft. dist., marked "X" in the snow blaze and BT in the base.

A Douglas fir, 26 ins. diam., bears N.34°W., 42.1 ft. dist., marked WC 1/4 S28 BT.

B.T.'s were signed and blazes painted red.

A steel fence post, with Land Survey Monument and Property Boundary decals attached, adjacent to monument.

Date of Work 5/22/2000 Mag. Decl. 18° Assumed

SURVEYOR'S CERTIFICATE

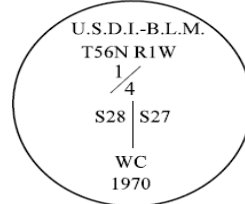
The fieldwork was performed and corner record prepared by me or under my direction in conformance with the Corner Perpetuation Act.

Name Frank S. Wratni
R.L.S. No. 4343
Employer U.S.D.A. Forest Service

SEAL, SIGNATURE, DATE:

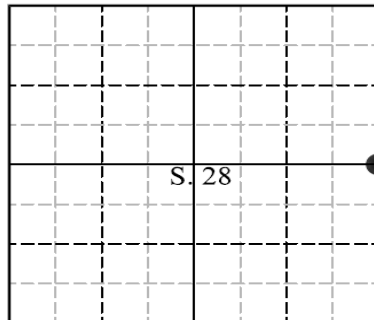


MARKS ON MONUMENT SET:



For County Use Only

DIAGRAM



SUBD. INDEX NO. (if applicable) _____

COUNTY Bonner

SECTION S.28

56N1W T13 0522 00

IPNF - Kaniksu N.F.

TOWNSHIP 56N

RANGE 1W

BOISE MERIDIAN

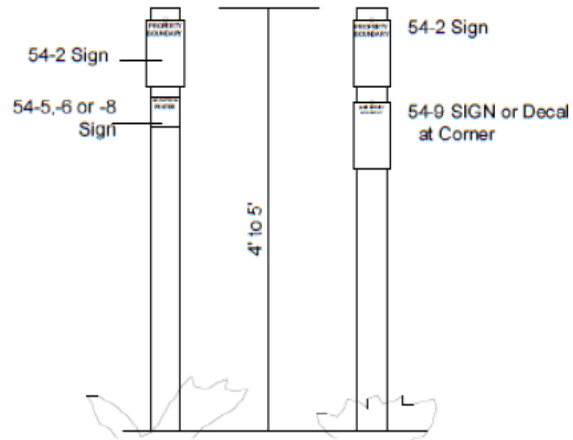
CROSS INDEX T13

Intermountain Region 4
BOUNDARY MANAGEMENT
CONTRACT SPECIFICATIONS

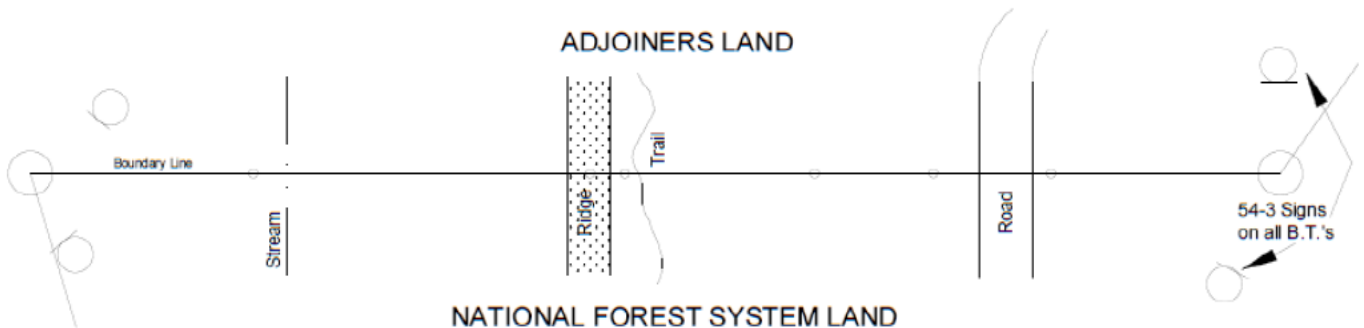
EXHIBIT .521 - SIGN PLACEMENT

NOTES:

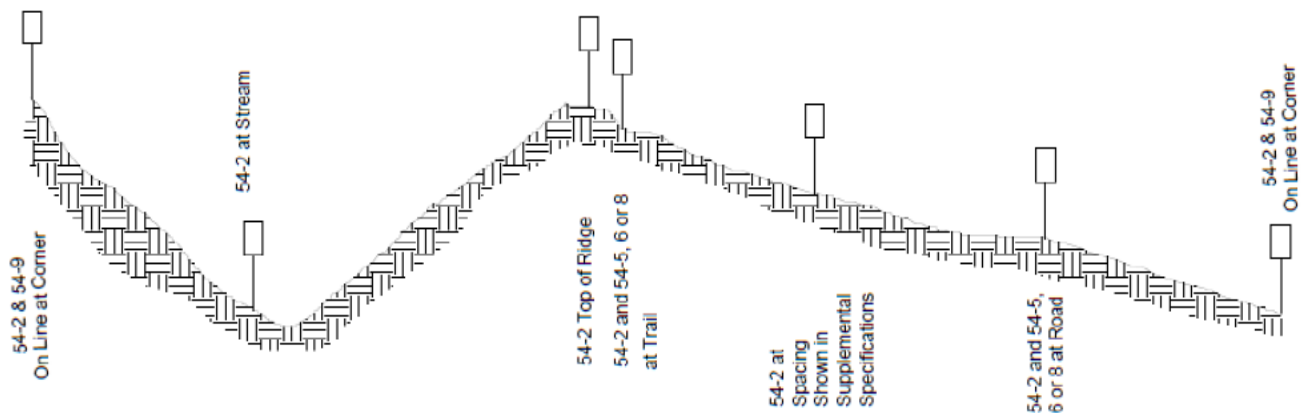
1. Posts shall be plumb and solidly planted with a minimum of 18" in the ground or supported to 24" above the base with a rock mound.
2. Posts with attached decal shall be set per manufacturers recommendations.
3. Sign 54-5, 54-6, or 54-8, "Location Poster," shall be placed where all marked boundary line cross roads and trails.
4. Post shall be set on the cut side of road, off the maintained surface.



**TYPICAL SIGN
AND POST INSTALLATION**



EXAMPLE PLAN VIEW OF BOUNDARY LINE SHOWING SIGN PLACEMENT

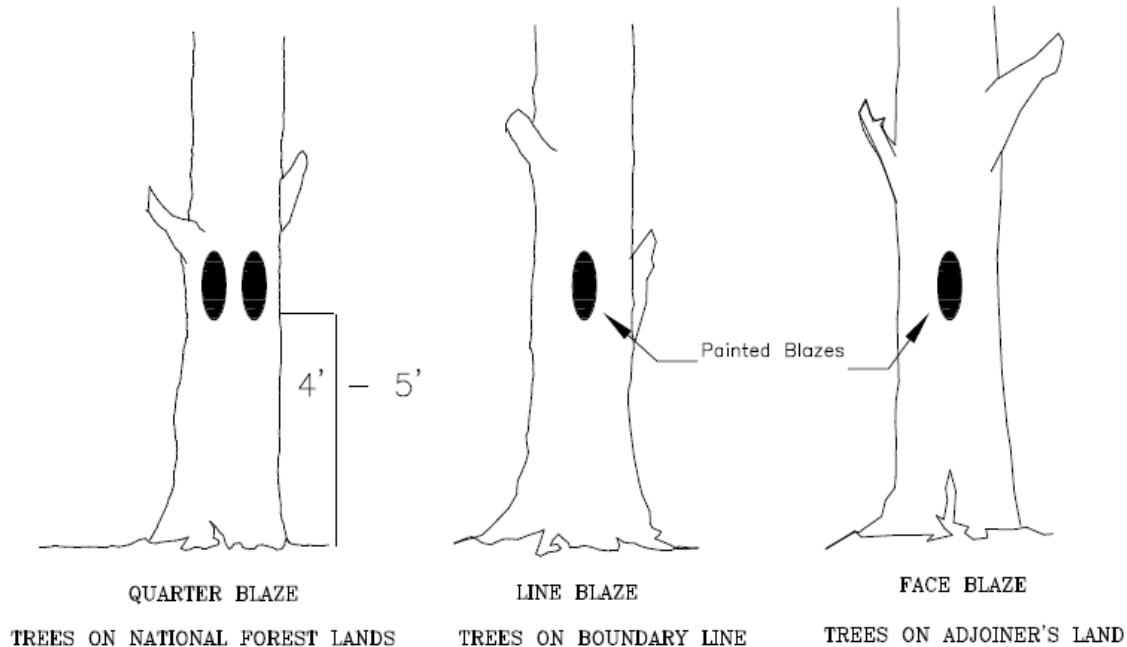


EXAMPLE PROFILE OF BOUNDARY LINE SHOWING SIGN PLACEMENT

Intermountain Region 4
BOUNDARY MANAGEMENT
CONTRACT SPECIFICATIONS

EXHIBIT .522 - MARKING AND POSTING THE BOUNDARY LINE

GENERAL APPEARANCE



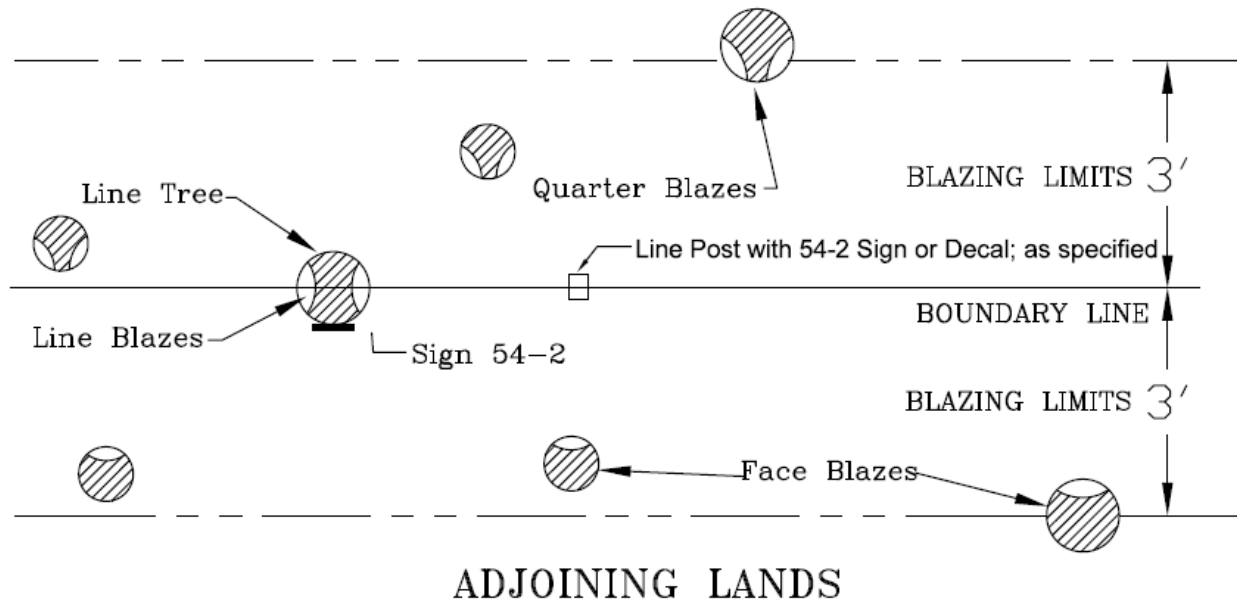
ORIENTATION OF BLAZES

NOTES

1. The three types of markings used on trees along the National Forest boundary lines are referred to as Quarter Blazes, Face Blazes and Line Blazes. The markings are used at the locations shown on page 2 of this Exhibit.
2. A blaze is made by cutting off a vertical strip of bark and a very thin layer of the underlying wood tissue. The strip shall be about 6 to 8 inches long, 2 to 4 inches wide, and the top and bottom ends shall be smoothed out.
3. A durable, oil base, red enamel paint shall be neatly applied to all blazes.

EXHIBIT .522 - MARKING AND POSTING THE BOUNDARY LINE

NATIONAL FOREST SYSTEM LANDS

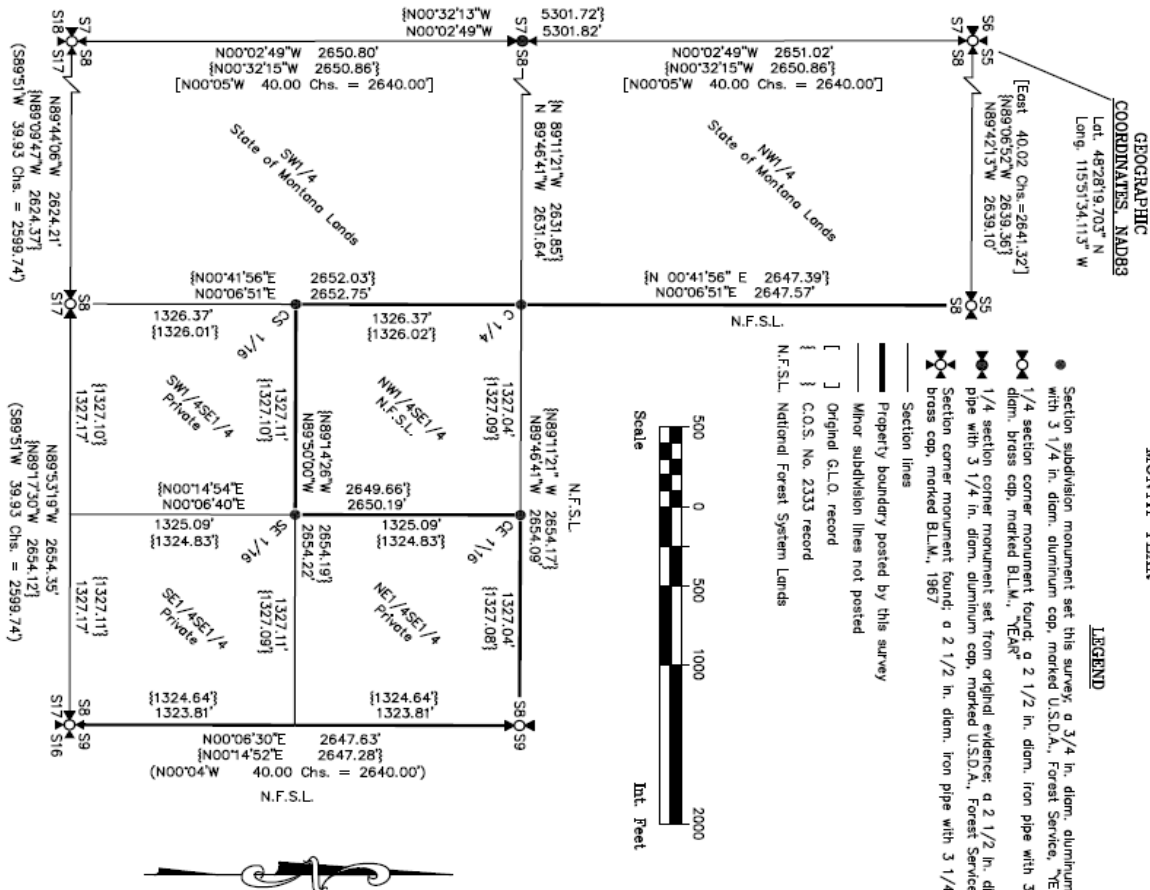


Notes:

1. Permission shall be obtained from adjoining landowners before their side of the boundary line is cleared or blazed.
2. Trees on National Forest System land within 3 feet of the boundary line shall be quarter blazed as shown.
3. Trees on adjoining land within 3 feet of the boundary line shall be face blazed as shown.
4. Trees on the boundary line shall be line blazed on both sides of the line tree along the direction of the line as shown above.
5. Posts, line monuments and line blazes shall be placed only on the boundary line after it has been located to the standard specified in the contract.
6. Decals 54-2 or 54-1 shall be placed on specified line posts only after the post has been thoroughly cleaned.
7. Sign 54-2 or 54-7 shall be attached to all line trees, at the spacing specified in the special project specifications, with 2 aluminum nails. The nails shall be left protruding at least 1/2 inch on trees to allow for growth.

Intermountain Region 4
BOUNDARY MANAGEMENT
CONTRACT SPECIFICATIONS

EXHIBIT .600 - EXAMPLE PLAT



FOREST SUPERVISOR'S CERTIFICATE

This survey was made at my request for National Forest Management purposes

Signature
NAME, Forest Supervisor

Date

PURPOSE OF SURVEY

The purpose of this survey is to retracement and mark the boundaries of existing parcels, distinguishing between National Forest System Lands and private lands, therefore no division of land is hereby created, exempting this survey from subdivision review, Sec. 76-3-207, MCA.

HISTORY OF SURVEY

1893, Original G.L.O. Township and Subdivision by Daniel P. Munhroe
1967, B.L.M. remonumentation of G.L.O. corners by Woodrow E. Damm
1995, C.O.S. No. 2333, subdivided Sec. 8 by John N. Good, Reg. No. 0005

METHOD OF SURVEY

Project control was accomplished using survey grade Global Positioning System observations with a constrained Least Squares Adjustment, based on HARN control stations "Freeman" and "Key".

A one second total station with data collector was utilized with closed traverse procedures to measure to original controlling corners and project control. Offsets to true line were computed for marking and posting the property line. The Property Boundary marking and posting was done by blazoning and painting trees, and placing galvanized steel delineator posts with Property Boundary decal, along the line at approximately 250 foot intervals.

BASIS OF BEARING

Bearings are based upon the geodetic meridian from GPS ties to the HARN network. A local Transverse mercator coordinate system was established with the origin at the northwest corner of Secs. 8, where the geodetic forward bearing equals the local grid bearing.

SURVEYOR'S NOTE

Additional monument information can be found on the Certified Corner Recordation documents filed DATE 00/00/00 on assigned pages "0000" through "0011" at the APPROPRIATE County.

SURVEYOR'S CERTIFICATE:

Signature
John Q. Surveyor, "STATE" Lic. No. 2001LS

Date

CERTIFICATE OF EXAMINING COUNTY OFFICIAL:

Approved this _____ day of _____, 200__ A.D.

Signature

Examining Official

CERTIFICATE OF CLERK AND RECORDER:

State of "APPROPRIATE STATE", County of "PROPER COUNTY", filed this _____ day of _____, 200__ A.D. at _____ o'clock

Signature

Signature

County Clerk Recorder

DOCUMENT NUMBER:

CERTIFICATE OF SURVEY NO.

Assigned by County
Assigned by County

Intermountain Region 4
BOUNDARY MANAGEMENT
CONTRACT SPECIFICATIONS

EXHIBIT .620 - CADASTRAL PLAT CHECKLIST

This list shall be checked, signed and dated by a reviewer through all phases of preparing the record.
A copy shall be sent with the draft plat submitted for review.

T. _____, R. _____, Sec.(s) _____,

_____ Mer., _____

_____ National Forest Project _____

Reviewer

Date

Plat Contents

- _____ Sheet Size
- _____ Border Size
- _____ Surveyed Lines Solid
- _____ Unsurveyed Lines Dashed
- _____ Line Weights Adequate
- _____ Lettering Legible
- _____ Title conforms to State law
- _____ Subtitle reflects type of survey
- _____ Legal Identifier (Section, Township, Range, Meridian, MS,

HES)

Orientation

- _____ Scale Bar
- _____ North Arrow
- _____ Geographic Position (1 minimum, 2 or more if project extents over 1.5 miles)
- _____ Datum and Method used to determine
- _____ Tie to Geodetic Control shown
- _____ Name and Establishment Date of Control Station(s) used

Plat Memo

- _____ History of Surveys
 - _____ Townships
 - _____ Sections
 - _____ Subdivisions
 - _____ Other
- _____ Purpose of Survey
 - _____ To Mark and Post National Forest Property Boundary
 - _____ Reference to type of posts set and intervals.

Intermountain Region 4
BOUNDARY MANAGEMENT
CONTRACT SPECIFICATIONS

EXHIBIT .620 - CADASTRAL PLAT CHECKLIST

- ☐ Date of Survey
- ☐ Project Name / Contract No.
- ☐ Basis of Bearing
- ☐ Data available in other sources
- ☐ Methods
- ☐ Instrumentation
- ☐ Positional Accuracy Statement
- ☐ Special Circumstances
- ☐ Disclaimer (Representation of Unsurveyed Features)

The Survey

- ☐ Records units of measure shown
- ☐ Reported measurements (proper precision)
- ☐ Corner Status evident
- ☐ Legend complete
 - ☐ Reflects corner status
 - ☐ Distinguishes Posted National Forest Property Boundary
- ☐ Adequate Information to determine and check survey methods
- ☐ On Mineral Survey or HES, tie to Rectangular Survey shown
- ☐ On MS, HES, etc, rectangular lines projected where feasible
- ☐ Ownership Labeled
 - ☐ "Private Lands" and "National Forest System Lands"
- ☐ Aliquot parts, lots, labeled
- ☐ Acreage properly shown
- ☐ Topography shown and labeled
- ☐ Cultural Features, fences, etc. adjacent to line shown and labeled

Certifications

- ☐ Surveyor's Certificate
- ☐ Forest Supervisor's Certificate
- ☐ County Certification/Approval/Recording Information

Comments: _____

Intermountain Region 4
BOUNDARY MANAGEMENT
CONTRACT SPECIFICATIONS

EXHIBIT .910 - EXAMPLE SURVEYOR'S REPORT

CADASTRAL SURVEY REPORT

_____ LAND SURVEY

_____ CONTRACT No.

SECTION(s) _____

T(s) _____, R(s) _____, _____ MERIDIAN

_____ RANGER DISTRICT

_____ NATIONAL FOREST

_____ COUNTY

_____ STATE

CONTRACTOR

_____ NAME

_____ ADDRESS

_____ CITY

_____ STATE

_____ ZIP

AWARDING OFFICE

_____ NATIONAL FOREST

_____ ADDRESS

_____ CITY

_____ STATE

_____ ZIP

CONTRACTING OFFICER: _____

CONTRACTING OFFICER'S REPRESENTATIVE: _____

INSPECTOR(s) _____

Page 1 of 3

*Intermountain Region 4
BOUNDARY MANAGEMENT
CONTRACT SPECIFICATIONS*

EXHIBIT .910 - EXAMPLE SURVEYOR'S REPORT

PURPOSE: The purpose of this contract was to perform professional land surveying services.
This contract included:

(List each required service, which will normally correspond to the contract pay items)

HISTORY OF PRIVATE SURVEYS: Briefly state the name and date of any private surveys, that are not indicated on the cadastral plat, that affect this survey. The report should detail why the private survey was or was not used in the conduct of this survey.

PROCEDURE:

CORNER SEARCH CONDUCTED: State the result of corner search efforts. It is not necessary to repeat the detailed information contained in the corner record forms, but the minimum information required is (1) which corners were located by original physical evidence, (2) which corners were located by parole evidence, (3) other surveyor's monuments were accepted, and (4) which corners were considered lost and a brief explanation of the extent of search efforts for the original physical evidence of the monument and accessories.

REMONUMENTATION ACCOMPLISHED: State which original corners were remonumented.

CONTROL SURVEY: Briefly describe the method or combination of methods that were used to (1) obtain the position of the controlling corners,(2) layout the position of lost corners and any new corners required to be established by the contract, and (3) locate the position of points on the boundary necessary for marking and posting the line. An example follows:

"A 5" total station with specifications of plus or minus 2mm plus 2 PPM was used on the control survey. The instrument was in proper adjustment and a base line user's report for the EDM is attached. The control survey utilized traverse methods that formed closed loops, which were adjusted by least squares method. Boundary lines were located by running true line between corner monuments. Angles were taken one direct and one reverse and these two angles were averaged and used for computations. Backsights were made to nails in the top of 1-meter laths and foresights were to plumb range poles. The estimated sighting error is 0.03 feet average. Distances were measured once to the foresight leg. Reductions to the horizontal were accomplished by means of vertical angles repeated twice in the direct and reverse position to the center of the prism, which was at a constant height of 6 feet. Computed moves were accomplished by turning the angle and shooting the distance direct and reverse.

*Intermountain Region 4
BOUNDARY MANAGEMENT
CONTRACT SPECIFICATIONS*

EXHIBIT .910 - EXAMPLE SURVEYOR'S REPORT

Monumentation: Explain the rationale for accepting or rejecting any other survey monuments found in the vicinity of a computed corner position. Explain what proportioning method was used to reestablish lost corners from adjacent existing corners. List which corners were established and monumented.

LAND OWNER CONTACTS AND CONFLICTS: Cite the result of landowner contact for permission to enter and mark and the private side of the boundary line. Include a brief statement of any apparent conflicts of possession or title discovered.

ENCROACHMENTS: Include a narrative of encroachments discovered and reported to the Contracting Officer Representative.

DATE OF SURVEY:

Survey Commenced _____

Survey Completed _____

PERSONNEL:

Contractor's Representative _____

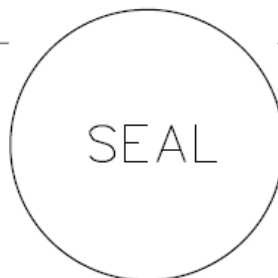
Land Surveyor in charge _____

Assistants: _____

SURVEYOR'S CERTIFICATE: I, _____, (State) Registered Land Surveyor No. _____ do hereby certify that this report is a true and correct account of a survey made by me or under my direction, under U.S.D.A. Forest Service Contract No. _____.

Signature

Date



Intermountain Region 4
BOUNDARY MANAGEMENT
CONTRACT SPECIFICATIONS

EXHIBIT .962 - RIGHT-OF-WAY SURVEY CHECKLIST

USDA-FOREST SERVICE

RIGHT-OF-WAY CHECKLIST
(REF. FSH 7109.15, R-1 25.65)

Project _____ Number: _____

Ownership _____

INSTRUCTIONS: Use this form as an aid in preparation and checking of right-of-way plats. All questions should be answered by placing an "X" in the box, or if the question is not applicable, indicate by drawing a line through the box.

1. Is the following information shown in the title block of the plat?

A. U.S. Department of Agriculture	H. Name of State
B. Forest Service	I. Name and Number of Project
C. Northern Region	J. Name of Surveyor
D. Name of Forest	K. Date of Survey
E. Right-of-Way Plat	L. Scale of Plat (Use Bar Scale)
F. Description of Land Crossed	M. Date of Plat
G. Name of County	

☐ YES
2. If there is more than one sheet, is there an index map showing the relation of the right-of-way to the found corners and PLS subdivisions, mineral survey, or homestead entry survey? Is each sheet numbered? ☐ YES
3. Is a north arrow shown on each sheet of the plat? ☐ YES
4. Is the origin of bearing shown? ☐ YES
5. Has the survey been closed? ☐ YES
6. Are the legal corners labeled as found and tied or, if not found, as projected?
Are the found corners adequately described? ☐ YES
7. Are the property lines marked projected where corners are not found, or if only one corner along a line is found? ☐ YES
8. Are the property lines correctly labeled with PLS record bearings and distances in record dimensions, and found Forest Service bearings and distances? ☐ YES
9. Are all pertinent occupancy lines including existing right-of-way lines correctly labeled? ☐ YES
10. Is the number of corner ties shown commensurate with the value of the land and ROW improvements? ☐ YES
11. Are ties to found corners shown and properly labeled at each change of ownership? ☐ YES
12. Is the station and the bearing of the tangent to the P.O.C. shown where there is a change of ownership on a curve? ☐ YES

R1-FS-7154-1 (07/83)

Intermountain Region 4
BOUNDARY MANAGEMENT
CONTRACT SPECIFICATIONS

EXHIBIT .962 - RIGHT OF WAY SURVEY CHECKLIST

- | | |
|---|------------------------------|
| 13. Are the L and P lines properly labeled? | ☐ YES |
| 14. Are existing natural monuments and cultural improvements shown, with ties when appropriate? | ☐ YES |
| 15. Are appropriate subdivisions and/or Government lots shown and correctly labeled? | ☐ YES |
| 16. Are the full names of the property owners shown on all subdivisions and other parcels of land crossed or closely approached by the right-of-way? Are corporate landowner's names legally correct? | ☐ YES |
| 17. Is the right-of-way width dimensioned on each sheet and at each change of width, and do the dimensions tie the right-of-way to the centerline. | ☐ YES |
| 18. Is the right-of-way acreage properly shown? | ☐ YES |
| 19. Have all the survey data and survey computations been checked? | ☐ YES |
| 20. Has all plotting been checked? | ☐ YES |
| 21. Is the ROW plat approved and sealed by an RLS of the jurisdiction? | ☐ YES |
| 22. Is the ROW plat approved by the Grasslands Engineer? | ☐ YES |
| 23. Has the plat been checked for copying errors and omissions and have all bearings and distances been checked with protractors and scale? | ☐ YES |
| 24. Is the plat legible, neat, and can it be readily folded to 8-1/2 x 11 inch size for filing? Have you considered reducing the plat photographically to letter size? | ☐ YES |
| 25. Is the Beginning and Ending ROW shown? | ☐ YES |

CHECKED BY:

APPROVED BY:

Signature _____
Date _____

Signature _____
Date _____